Review of Economic Studies (2026) **93**, 2284–2312 https://doi.org/10.1093/restud/rdaf087
© The Author(s) 2025. Published by Oxford University Press on behalf of The Review of Economic Studies Limited.
This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial License (https://creativecommons.org/licenses/by-nc/4.0/), which permits non-commercial re-use, distribution, and reproduction in any medium, provided the original work is properly cited. For commercial re-use, please contact reprints@oup.com for reprints and translation rights for reprints. All other permissions can be obtained through our RightsLink service via the Permissions link on the article page on our site—for further information please contact journals.permissions@oup.com.
Advance access publication 30 September 2025

The Surrogate Index: Combining Short-Term Proxies to Estimate Long-Term Treatment Effects More Rapidly and Precisely

Susan Athey
Graduate School of Business, Stanford University and NBER, USA
Raj Chetty
NBER and Department of Economics, Harvard University, USA
Guido W. Imbens
Graduate School of Business, and Department of Economics, Stanford University, USA
and
Hyunseung Kang
Department of Statistics, University of Wisconsin at Madison, USA

First version received June 2020*; Accepted August* 2024 *(Eds.)*

A common challenge in estimating the impact of interventions (*e.g.* job training programmes, educational programmes) is that many outcomes of interest (*e.g.* lifetime earnings or other labour market outcomes) are observed with a long delay. In biomedical settings, this is often addressed by using short-term outcomes as so-called "surrogates" for the outcome of interest, *e.g.* tumour size as a surrogate for mortality in cancer studies. We build on this literature by combining multiple, possibly qualitatively distinct, short-term outcomes (*e.g.* short-run earnings and employment indicators) systematically into a "surrogate index". Under the Prentice surrogacy assumption, which requires that the primary outcome is independent of the treatment conditional on the surrogates, we show that the average treatment effect on the surrogate index equals the treatment effect on the long-term outcome. We also relate the surrogacy assumption to a set of structural, causal assumptions. We then characterize the bias that arises from violations of each of the key assumptions, and we provide simple methods to validate these assumptions using additional observed outcomes. We apply our method to analyse the long-term impacts of a multi-site job training experiment in California. Rather than waiting a full 9 years to directly observe the long-term impact, we show that it is possible to use short-term (the first six quarters) outcomes as surrogates. Given the surrogacy assumption one could have estimated the programme's long-term impacts on mean employment rates using the employment rates observed in the first six quarters, with a 35% reduction in standard errors relative to a simple difference in means estimator based on all 9 years of data.

Key words: Potential outcomes, Causality, Surrogate outcomes, Surrogate score, Surrogate index, Mediators, Propensity score, Principal stratification, Job training

JEL codes: C40

The editor in charge of this paper was Francesca Molinari.

Downloaded from academic.oup.com/restud/article/93/4/2284/8268796 by guest on 06 November 2026

1. INTRODUCTION

A fundamental challenge for evaluating interventions is that the primary outcomes of interest are often hard to measure. For example, researchers are often interested in the effect of the policy on some long-term outcome but do not observe that in their study. Instead, they observe a number of short-term outcomes that are all related to this primary outcome of interest.

One setting where this type of problem arises involves an educational policymaker evaluating a policy that would change class size. The ultimate goal may be to improve long-term labour market outcomes for the students. However, the decision regarding the class size policy needs to be made at a time when only short-term outcomes such as test scores or other educational achievement measures are available. Another setting involves policymakers considering labour market interventions such as job search assistance or human capital acquisition programmes, where they may be primarily interested in the long-term labour market attachment of the participants, but in the short run they may only have access to outcomes such as employment records or earnings over a short period of time. In randomized experiments for medical interventions, the ultimate outcome of interest is often survival or quality-adjusted years of life. Survival rates may be high in the short run, and so typically such trials are evaluated in terms of surrogate measures, such as including tumour size or other measures of the progression of the disease, which can be measured earlier. In all of these types of setting, to make a timely decision, the policymaker needs to assess the programmes based on short-term outcomes. These challenges also arise in business settings. In the context of experimentation in digital technology companies, a discussion of the most important challenges ranks as the top concern that "While most experiments in the industry run for 2 weeks or less, we are really interested in detecting the long-term effect of a change. How do long-term effects differ from short-term outcomes? How can we accurately measure those long- term factors without having to wait a long time in every case?" (Gupta *et al.*, 2019, p. 21).

In these and many other examples, the researcher is faced with making recommendations regarding the future implementation of the intervention on the basis of measurements of its effect on a variety of sometimes disparate and possibly conflicting outcome measures. A key question is how to balance these different outcomes when making an overall assessment. In practice, researchers often deemphasize short-term outcomes for which they do not find statistically significant effects, instead making perhaps somewhat *ad hoc* qualitative assessments regarding the relative importance of the remaining short-term outcomes.

In this paper, we lay out a framework for analysing these issues. We consider the scenario in which researchers do not measure the primary outcome in the context of data containing information on the intervention. Instead, we assume that the researcher has a second, observational, dataset where the researcher observes the surrogates and the primary outcome but does not observe the treatment. In both samples, the researcher may also observe variables not affected by the treatment, such as pre-treatment characteristics of the participants.

We make four main contributions. First, we articulate three key assumptions under which the average effect of the treatment on the primary outcome is identified from the combination of the experimental and observational samples: (1) a standard assumption that the assignment in the experimental sample is *Unconfounded*; (2) a *Surrogacy* assumption which requires that the causal path from the treatment to the primary outcome goes through the surrogates (Prentice, 1989; Day and Duffy, 1996; Begg and Leung, 2000; Frangakis and Rubin, 2002); and (3) a *Comparability* or external validity assumption, which requires that the observational and

Downloaded from academic.oup.com/restud/article/93/4/2284/8268796 by guest on 06 November 2026

experimental samples are comparable in the sense that the outcome distributions conditional on surrogates and pre-treatment variables are identical.

Under these three assumptions, the average effect of the treatment on the primary outcome can be estimated as the average effect of the treatment on an aggregate of the surrogates, which we label the surrogate index. This index combines the individual surrogates through their predicted value of the primary outcome. For example, when studying the impact of class size, the primary outcome might be high school graduation, while the two surrogates might be mathematics and reading scores. For the special case of linear models, the proposal boils down to multiplying the causal effects of the intervention on the two scores (which can be estimated in the experimental sample) by the coefficients from a linear regression of the primary outcome on the two scores in the observational sample. The approach replaces a subjective assessment of the relative importance of the two short-term measures by an objective data-driven criterion, namely the predictive power of the scores for the outcome of interest.

In our second contribution, we derive the efficiency bound and propose various efficient estimators under various scenarios, including scenarios with a single sample or two samples, as well as with and without Surrogacy. This allows us to quantify the information content of the Surrogacy assumption.[1]

In our third contribution, we provide bounds on the biases that arise in scenarios where either or both of Surrogacy or Comparability are violated. We show that even if these assumptions fail to hold (but unconfoundedness does hold), the proposed estimators still estimate a well-defined causal effect, by providing a principled way of combining short-term outcomes in a single measure through their predicted effect on the long-term outcome.

In our fourth contribution, we evaluate these methods in the context of a labour market programme where we observe long-term (thirty-six quarters) outcomes in four locations. Following an approach popularized by LaLonde (1986), we put aside part of the data and investigate whether we could have estimated the long-term effects without having long-term experimental data. Specifically, we take one of the locations, Riverside, and put aside the long-term outcome for individuals from that location. Then we take the other three locations, Alameda, Los Angeles, and San Diego, and put aside the treatment assignment for that sample. We investigate whether these two samples allow us to recover the experimental long-term effects in Riverside using surrogates corresponding to the first T quarters of outcomes (employment, earnings, and aid indicators). We find that combining six quarters of outcome data into a surrogate index suffices to obtain estimates close to the long-run effects. Using the additional data that were put aside for the main analysis, we also directly test whether the critical assumptions, Surrogacy and Comparability, hold given various alternative sets of surrogates.

We recognize that the credibility of the Surrogacy assumption may be questioned in any given application, especially when viewed in isolation. Therefore, we view the best path forward as building a "library" of surrogate indices in which researchers systematically catalogue across several studies the smallest set of surrogates that successfully match long-term outcomes of interest (*e.g.* earnings, mortality, and educational attainment). If one establishes, for instance, that six quarters of employment and earnings data are sufficient to predict the impacts of many different job training programmes—as our cross-site comparisons of the Greater Avenues to Independence (GAIN) programme suggest—then the long-term impacts of future job training programmes could be credibly estimated using the established six-quarter surrogate index. We view the empirical application in this paper as providing one element of such a library and hope

1. We are grateful to Kevin Chen and David Ritzwoller for pointing out an error in one of our earlier efficiency bound calculations. See Chen and Ritzwoller (2023) for more details.

Downloaded from academic.oup.com/restud/article/93/4/2284/8268796 by guest on 06 November 2026

future work will expand upon it by identifying surrogate indices that match estimated long-term impacts in other applications.

This study is related to three main bodies of literature, surrogacy, mediation, and missing data. We extend the literature on surrogacy (Prentice, 1989; Day and Duffy, 1996; Fleming and DeMets, 1996; Begg and Leung, 2000; Xu and Zeger, 2001; Lauritzen, 2004; Alonso *et al.*, 2006; D'Agostino *et al.*, 2006; Qu and Case, 2006; Weir and Walley, 2006; Gilbert and Hudgens, 2008) by formally including the presence of a second observational sample that is used to estimate the relationship between surrogates and the primary outcome and articulating the assumptions that justify doing so. In doing so, we allow for uncertainty in the estimation of this surrogates/outcome relationship, whereas the previous literature took this relation as known. We also consider biases arising from violations of Surrogacy and Comparability.

In addition, this study builds on the literature on mediation (Baron and Kenny, 1986; van der Laan and Petersen, 2004; Imai *et al.*, 2010; Zheng and van der Laan, 2012; Tchetgen Tchetgen and Shpitser, 2014; VanderWeele, 2015), which considers the decomposition of an average treatment effect (ATE) into the direct effect of a treatment on an outcome and indirect effects that flow through a mediator. In the mediation setup, all three key variables—the outcome, the treatment, and the mediator—are observed for the same units. The goal in the mediation literature is to determine the relative magnitudes of the direct and indirect effects. In our surrogacy analysis, we focus on the case in which the direct effect is absent by assumption.

This paper is also related to the classical missing data literature in statistics (Rubin, 1976, 2004; Little and Rubin, 2014). Our key assumptions are closely related to the Missing At Random (MAR) assumption. Our approach can be viewed as a special case of approaches that combine data sets, *e.g.* Ridder and Moffitt (2007) and Chen *et al.* (2008). In particular, Rässler (2004, 2012) refers to our setting, where one variable is missing in one part of a sample and a second variable missing in the remainder of the sample, as a "data fusion" setting. Graham *et al.* (2016) discuss efficient estimation for a particular set of models defined by moment conditions in such a data fusion setting, where they allow the treatment to be a general random variable, rather than a binary indicator as in our setup.

The paper is organized as follows. Section 2 sets up the problem and introduces the notation. Section 3 discusses the critical assumptions and links the setup to the mediation and missing data literature. Section 4 discusses identification and the efficiency bounds. Section 5 presents formulas for bias when the surrogacy assumption fails and derives bounds on the degree of bias. Section 6 discusses estimation. Section 7 presents the empirical application. Section 8 concludes.

2. SETUP AND NOTATION

We define two samples, an Experimental (E) sample and an Observational (O) sample, with N_{E} and N_{O} units or individuals, respectively. It is convenient to view the data as consisting of a single sample of size $N = N_{\mathrm{E}} + N_{\mathrm{O}}$, with $P_i \in \{\mathrm{O}, \mathrm{E}\}$ a binary indicator denoting the sample to which unit i belongs.

For each unit, there is a binary treatment of interest, $W_i \in \{0, 1\}$, and a scalar primary outcome, denoted by Y_i. This outcome is not observed for individuals in the experimental sample. In addition, there are intermediate or secondary outcomes, which we refer to as surrogates (to be defined precisely in Section 3.2), denoted by S_i for each unit. Typically, the surrogate outcomes are vector-valued in order to make the properties we define plausible. Finally, we measure pre-treatment covariates X_i for each unit, known not to be affected by the treatment.

Following the potential outcomes framework or Rubin Causal Model (Rubin, 1974; Holland, 1986; Imbens and Rubin, 2015), individuals in this group have two pairs of potential outcomes:

Downloaded from academic.oup.com/restud/article/93/4/2284/8268796 by guest on 06 November 2026

TABLE 1
Observation scheme: ✓ is observed, ? is missing

	Long-term				Pre-treatment
Units	Sample P_i	Treatment W_i	Outcome Y_i	Surrogate S_i	Variables X_i
1 to $N_{\rm E}$	E	✓	?	✓	✓
$N_{\rm E}+1$ to $N_{\rm E}+N_{\rm O}$	O	?	✓	✓	✓

$(Y_i(0), Y_i(1))$ and $(S_i(0), S_i(1))$. The realized outcomes are related to their respective potential outcomes as follows:

$$Y_i \equiv Y_i(W_i) = \begin{cases} Y_i(0) \text{ if } W_i = 0, \\ Y_i(1) \text{ if } W_i = 1, \end{cases} \text{ and } S_i \equiv S_i(W_i) = \begin{cases} S_i(0) \text{ if } W_i = 0, \\ S_i(1) \text{ if } W_i = 1. \end{cases}$$

Overall, the units are characterized by the values of the septuple $(Y_i(0), Y_i(1), S_i(0), S_i(1), X_i, W_i, P_i)$. We do not observe the full septuple for any units. Rather, for units in the experimental sample, we observe the triple (X_i, W_i, S_i) with support $(\mathbb{X}, \mathbb{W}, \mathbb{S})$ where $\mathbb{W} = \{0, 1\}$. In the observational sample, we do not observe to which treatment each of the $N_{\rm O}$ individuals was assigned. We observe the triple (X_i, S_i, Y_i), with support $\mathbb{X}$, $\mathbb{S}$, and $\mathbb{Y}$, respectively. To simplify the exposition, we analyse the data as if we have a random sample from a population of units for which we observe the quintuple $(P_i, X_i, S_i, \mathbf{1}_{P_i={\rm E}}W_i, \mathbf{1}_{P_i={\rm O}}Y_i)$, where we treat P_i as a random variable taking on the values {O, E}.

Assumption 1. We have a single random sample of size N drawn from the joint distribution of $(P_i, X_i, S_i, W_i, Y_i)$, where we observe for each unit in the sample $(P_i, X_i, S_i, \mathbf{1}_{P_i={\rm E}}W_i, \mathbf{1}_{P_i={\rm O}}Y_i)$.

We summarize this data setup in Table 1. The setup differs from those in Athey *et al.* (2020) and Kallus and Mao (2025), where we would also observe the treatment in the observational sample, but in the experimental sample we would still not observe the primary outcome.

We are interested in the ATE on the primary outcome in the population from which the experimental sample is drawn:

$$\tau \equiv \mathbb{E}[Y_i(1) - Y_i(0) \mid P_i = {\rm E}]. \tag{1}$$

The same issues we study in the current paper apply to other estimands, such as the average treatment effect for the treated units, or the average for the observational sample.

An implicit assumption in our setup is that the two variables that are common to both samples, S_i and X_i, measure the same underlying variables in both samples. In some cases, it is possible that in one of the two samples, a coarser version is measured, for example, age or education may be measured in multi-year categories rather than in years. In that case, a simple solution is to proceed by using the coarser version of the variables as corresponding to the surrogate of pre-treatment variable. Another complication arises if the unit of observation differs in two samples, say individuals versus zipcodes. Again additional assumptions are required to link the variables between samples.

Table 2 summarizes key definitions and notation.

Downloaded from academic.oup.com/restud/article/93/4/2284/8268796 by guest on 06 November 2026

TABLE 2
Notation and definitions

Sampling indicator	$P_i \in \{\text{E}, \text{O}\}$
Potential outcomes for primary outcome	$Y_i(0), Y_i(1)$
Potential outcomes for surrogates	$S_i(0), S_i(1)$
Binary treatment indicator	$W_i \in \{0, 1\}$
Realized value for outcome	$Y_i = Y_i(W_i)$
Realized value for surrogate	$S_i = S_i(W_i)$
Estimand	$\tau \equiv \mathbb{E}[Y_i(1) - Y_i(0) \mid P_i = \text{E}]$
	$\mu(s, w, x, p) \equiv \mathbb{E}[Y_i \mid S_i = s, W_i = w, X_i = x, P_i = p]$
Surrogate index	$\mu(s, x, p) \equiv \mathbb{E}[Y_i \mid S_i = s, X_i = x, P_i = p]$
	$\mu(w, x) \equiv \mathbb{E}[\mu(S_i, X_i, O) \mid W_i = w, X_i = x, P_i = E]$
	$\sigma^2(s, w, x, p) \equiv \mathbb{V}(Y_i \mid S_i = s, W_i = w, X_i = x, P_i = p)$
	$\sigma^2(s, x, p) \equiv \mathbb{V}(Y_i \mid S_i = s, X_i = x, P_i = p)$
	$\sigma^2(w, x) \equiv \mathbb{V}(Y_i \mid W_i = w, X_i = x, P_i = \text{O})$
Surrogate score	$\rho(s, x) \equiv \text{pr}(W_i = 1 \mid S_i = s, X_i = x, P_i = \text{E})$
Propensity score	$\rho(x) \equiv \text{pr}(W_i = 1 \mid X_i = x, P_i = \text{E})$
	$\rho \equiv \text{pr}(W_i = 1 \mid P_i = \text{E})$
Sampling score	$\varphi(s, x) \equiv \text{pr}(P_i = \text{E} \mid S_i = s, X_i = x)$
	$\varphi(x) \equiv \text{pr}(P_i = \text{E} \mid X_i = x)$
	$\varphi \equiv \text{pr}(P_i = \text{E})$
Conditional distribution of surrogates	$\pi(s \mid w, x) \equiv f_{S_i \mid W_i, X_i, P_i}(s \mid w, x, \text{E})$
	$\pi(s \mid x) \equiv f_{S_i \mid X_i, P_i}(s \mid x, \text{E})$
Influence function	$\psi(y, s, w, x, p)$

Notes: This table summarizes the notation. Conditional expectations and variances of the outcome Y_i are denoted by $\mu(\cdot)$ and $\sigma^2(\cdot)$, respectively. Conditional probabilities of the treatment are denoted by $\rho(\cdot)$. Conditional probabilities of the sample are denoted by $\varphi(\cdot)$ The arguments of these functions can be both the surrogates S_i and the pre-treatment variables X_i, or just the pre-treatment variables X_i.

3. THE CRITICAL ASSUMPTIONS: UNCONFOUNDEDNESS, SURROGACY, AND COMPARABILITY

In this section, we discuss the three key assumptions that together allow us to combine the observational and experimental samples and estimate the causal effect of the treatment on the primary outcome, exploiting the presence of the surrogates. The first assumption is *Unconfoundedness* or *Ignorability*, common in the programme evaluation literature (Rosenbaum and Rubin, 1983b; Imbens and Rubin, 2015), which ensures that adjusting for pre-treatment variables leads to valid causal effects in the experimental sample. The second assumption is the *Surrogacy condition* due to Prentice (1989), that allows us to use the surrogate variables to proxy for the primary outcome. The third assumption is *Comparability*, which formalizes the connection between the two samples. This assumption is rarely stated formally, but plays an important role in our analysis.

3.1. *Unconfoundedness*

For the individuals in the experimental group, the propensity score is the conditional probability of receiving the treatment: $\rho(x) \equiv \text{pr}(W_i = 1 \mid X_i = x, P_i = \text{E})$. We assume that for individuals in the experimental group, treatment assignment is unconfounded, and we have overlap in the distribution of pre-treatment variables between the treatment and control groups (Rosenbaum and Rubin, 1983b; Imbens and Rubin, 2015):

Downloaded from academic.oup.com/restud/article/93/4/2284/8268796 by guest on 06 November 2026

Assumption 2 (UNCONFOUNDED TREATMENT ASSIGNMENT/STRONG IGNORABILITY)**.**

(i)

$$W_i \perp\!\!\!\perp (Y_i(0), Y_i(1), S_i(0), S_i(1)) \mid X_i, P_i = \mathrm{E},$$

(ii) $0 < \rho(x) < 1$ for all $x \in \mathbb{X}$.

This assumption, widely used in the causal inference literature, implies that in the experimental sample, we can estimate the average causal effect of the treatment on the surrogates by adjusting for pre-treatment variables. We would also have been able to estimate the causal effect on the primary outcome had the primary outcome been measured in the experimental sample. In many applications of surrogacy approaches, the treatment in the experimental sample is assigned completely randomly. In that case, this assumption is satisfied by design. However, unconfoundedness is all that is required.

3.2. *Surrogacy*

Next, we discuss the second critical assumption, surrogacy. We also introduce two concepts, the *surrogacy score*, similar to the propensity score, and the *surrogacy index*, to combine multiple surrogates.

3.2.1. The Prentice criterion. Prentice (1989) defines a surrogate as a post-treatment variable where conditioning on it makes the outcome and the treatment independent:

Assumption 3 (SURROGACY, PRENTICE CRITERION)**.**
(i)

$$W_i \perp\!\!\!\perp Y_i \mid S_i, X_i, P_i = \mathrm{E}.$$

and (ii) $0 < \rho(s, x) < 1$, for all $s \in \mathbb{S}, x \in \mathbb{X}$, and $0 < \mathrm{pr}(P_i = \mathrm{E}) < 1$.

Remark 1. If the quadruple (Y_i, S_i, W_i, X_i) were observed for all units, surrogacy would be a testable condition. With (S_i, W_i, X_i) observed for units in the experimental sample, and (Y_i, S_i, X_i) observed for units in the observational sample, this assumption has no testable implications.

Remark 2. Note that Surrogacy is formulated in terms of the realized outcome and surrogate values. In contrast, we formulated the ignorability condition (Assumption 2) in terms of the potential outcomes. This is partly to connect our discussion to the surrogacy literature (Prentice, 1989; Day and Duffy, 1996).

Surrogacy is often debated in empirical applications. Freedman *et al.* (1992) argue that the surrogate may not mediate the full effect of the treatment in many settings. For example, reductions in class size may affect earnings through changes in non-cognitive skills that are not fully captured by standardized test scores (Heckman *et al.*, 2006; Chetty *et al.*, 2011).

3.2.2. The surrogacy index and the surrogacy score. There are two scalar functions of the surrogates that play an important role in the analyses: the surrogate index and surrogate score.

Definition 1 (THE SURROGATE INDEX)**.** The surrogate index is the conditional expectation of the primary outcome given the surrogate outcomes and the pre-treatment variables, conditional on the sample:

Downloaded from academic.oup.com/restud/article/93/4/2284/8268796 by guest on 06 November 2026

$$\mu(s, x, p) \equiv \mathbb{E}\left[Y_i | S_i = s, X_i = x, P_i = p\right].$$

Remark 3. The surrogate index in the observational sample, $\mu(s, x, \text{O})$, is identified because we observe the triple (Y_i, S_i, X_i) in the observational sample.

Definition 2 (THE SURROGATE SCORE)**.** The surrogate score is the conditional probability of having received the treatment given the value for the surrogate outcomes and the covariates in the experimental sample:

$$\rho(s, x) \equiv \text{pr}(W_i = 1 \mid S_i = s, X_i = x, P_i = E).$$

The role the surrogacy score plays is similar to the role the propensity score plays in analyses under unconfoundedness (Rosenbaum and Rubin, 1983b). Here, if the surrogacy condition holds conditional on (S_i, X_i), it also holds conditional on the surrogacy score.

Proposition 1 (SURROGATE SCORE). *Suppose Surrogacy (Assumption 3) holds. Then:*

$$W_i \perp\!\!\!\perp Y_i \mid \rho(S_i, X_i), P_i = \text{E}.$$

All proofs are given in the Online Appendix.

3.2.3. The benefits of multiple surrogates. One theme of this paper is that having multiple short-term variables can make a surrogacy approach more plausible, the same way multiple pre-treatment variables can make the unconfoundedness assumption more plausible. Here, we discuss some illustrative examples.

The first example is illustrated in Figure 1A. Suppose the treatment is an educational intervention. This treatment affects the outcome of interest, some labour market outcome, *e.g.* earnings, through a number of different channels corresponding to different skill sets. These channels may include mathematics skills, language skills, and social skills. Using only one of these variables as a surrogate would lead to biased estimates because they would ignore the other causal paths. In this case, the set of three short-term variables collectively satisfy Unconfoundedness and Surrogacy.

The second case is illustrated in Figure 1B. In this setup, there is a variable, labelled "skills", that satisfies the critical assumptions for surrogacy. However, skills are not observed by the researcher. Instead, we have two noisy measures of this surrogate, say both a written and an oral exam. Collectively, these two variables may still not satisfy Surrogacy, since there may be impacts of skills on earnings not captured by the exams, but the bias from using both would be less than the bias from using only one candidate surrogate.

The third case is illustrated in Figure 1C. Here, there is a pathway from the treatment, an informative advertisement about an item, to the outcome, an indicator for the individual purchasing the advertised item, going through two variables that on their own could each serve as surrogates. These two variables are whether someone has interested in the item, and whether the individual engaged with the website where the item was sold. However, we only measure noisy versions of these surrogates. For the first surrogate, we observe whether an individual clicked on the advertisement for the item, and for the second surrogate, we observe the time spent on the website. Neither of these two observed variables is a valid surrogate, but the combination of the two generally removes more of the bias than a single one.

Figure 1D illustrates further the concerns with only using a single surrogate, the possibility of focusing on treatments that improve the surrogate variable but not the primary outcome. Suppose

Downloaded from academic.oup.com/restud/article/93/4/2284/8268796 by guest on 06 November 2026

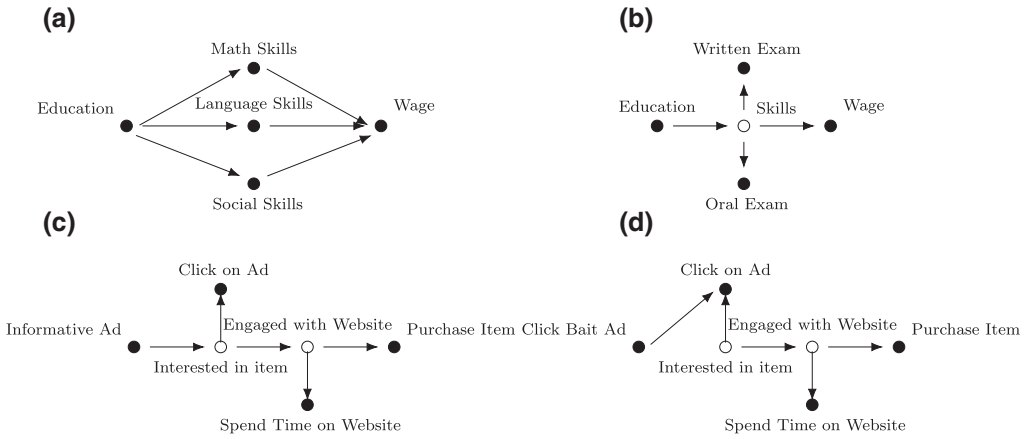


FIGURE 1

(a) Surrogacy assumption satisfied. (b) Multiple surrogates. (c) Multiple surrogates, Scenario 1. (d) Multiple surrogates, Scenario 2

that a researcher uses the single variable "click on ad" as a surrogate for the effect of the ad on purchases. If the observational sample was based on informative advertisements, there is likely a positive correlation between clicking on the advertisement and purchases. However, if the new treatment is uninformative, *e.g.* clickbait advertisement, with no effect on the actual interest in the item, the surrogacy analysis using click behaviour as the surrogate will be ineffective. Using both click behaviour and time spent on the website as surrogates will likely reduce the bias. The same argument implies that using multiple tests as surrogates can reduce problems with "teaching to the test", where the long-run impact of an intervention is not well captured by scores on a test.

3.3. *Comparability*

Surrogacy and Unconfoundedness by themselves are not sufficient for consistent estimation of τ because they do not place restrictions on how the relationship between Y_i and S_i in the observational sample compares to that in the experimental sample. As far as we know, such restrictions were not previously articulated in the surrogacy literature because the setup is typically one with just the separate experimental sample. However, a comparability assumption is implicit in the way the postulated relationship between the surrogate and the primary outcome is used in that literature. Related assumptions about the possibility of using causal estimates in one location to predict causal effects in a second location on the basis of distributions of pre-treatment variables are discussed in Hotz *et al.* (2005) and the literature on transportability, Pearl and Bareinboim (2014).

3.3.1. The comparability assumption. Let $\varphi \equiv \text{pr}(P_i = \text{E})$ be the probability of a unit being part of the experimental sample. We introduce the *Sampling Score,* the propensity to be in the experimental sample:

Definition 3 (SAMPLING SCORE)**.** The sampling score is $\varphi(s, x) \equiv \text{pr}(P_i = \text{E} \mid S_i = s, X_i = x)$.

Downloaded from academic.oup.com/restud/article/93/4/2284/8268796 by guest on 06 November 2026

The third key assumption we make is that the conditional distribution of Y_i given (S_i, X_i) in the observational sample is the same as the conditional distribution of Y_i given (S_i, X_i) in the experimental sample, and that the support of (S_i, X_i) in the experimental sample is a subset of that in the observational sample. Formally,

Assumption 4 (COMPARABILITY OF SAMPLES)**.**

(i) $P_i \perp\!\!\!\perp Y_i \mid S_i, X_i$,
(ii) $\varphi(s, x) < 1$ for all $s \in \mathbb{S}$ and $x \in \mathbb{X}$.

Similar to Unconfoundedness and Surrogacy, this is a strong assumption, but unlike those assumptions, it is rarely discussed explicitly. As we show in Section 5, by making it explicit we can discuss the biases arising from violations and improve the intuition when this assumption may be of concern. If the observational and experimental samples are substantially different in terms of the distribution of pre-treatment variables and surrogates, it would likely be more controversial to assume that, conditional on those variables, the outcome distributions are identical.

3.3.2. The surrogate index and the sampling score. We let $\mu(s, w, x, p)$ denote the conditional expectation of the primary outcome given pre-treatment variables, surrogates, treatment, and sample:

$$\mu(s, w, x, p) \equiv \mathbb{E}\left[Y_i \mid S_i = s, X_i = x, W_i = w, P_i = p \right]. \tag{2}$$

Comparability and Surrogacy together allow us to impute the missing primary outcomes in the experimental sample, as shown by the following proposition.

Proposition 2 (SURROGATE INDEX).

(i) *Suppose Assumption 3 (Surrogacy) holds. Then:*

$$\mu(s, w, x, \mathrm{E}) = \mu(s, x, \mathrm{E}), \quad \textit{for all } s \in \mathbb{S}, \ x \in \mathbb{X}, \quad \text{and} \quad w \in \mathbb{W}.$$

(ii) *Suppose Assumption 4 (Comparability) holds. Then:*

$$\mu(s, x, \mathrm{E}) = \mu(s, x, \mathrm{O}) \ \textit{for all } s \in \mathbb{S}, \quad \text{and} \quad x \in \mathbb{X}.$$

(iii) *Suppose Assumptions 3 (Surrogacy) and 4 (Comparability) hold. Then:*

$$\mu(s, w, x, \mathrm{E}) = \mu(s, x, \mathrm{O}) \ \textit{for all } s \in \mathbb{S}, x \in \mathbb{X}, \quad \text{and} \quad w \in \mathbb{W}.$$

Because we can estimate $\mu(s, x, \mathrm{O}) = \mathbb{E}[Y_i \mid S_i = s, X_i = x, P_i = \mathrm{O}]$, we can impute the missing Y_i in the experimental sample as $\mu(S_i, X_i, \mathrm{O})$.

3.4. *Surrogacy, mediation, instrumental variables, directed acyclical graphs, and missing data*

To provide context for the setup here and the key assumptions, it is useful to make a link to three related literatures, on mediation, instrumental variables, and missing data, respectively. We describe the causal structures for surrogacy, mediation, and instrumental variables using a directed acyclical graph (DAG) (Pearl, 2000). The interpretations provided in this subsection are not essential to the main results in the next section.

Downloaded from academic.oup.com/restud/article/93/4/2284/8268796 by guest on 06 November 2026

TABLE 3
Surrogacy versus mediation versus instrumental variables

	Surrogacy	Mediation	Instrumental Var
Left variable (L)	Treatment (W)	Treatment (W)	Instrument (Z)
Middle variable (M)	Surrogate (S)	Mediator (M)	Treatment (W)
Right variable (R)	Outcome (Y)	Outcome (Y)	Outcome (Y)
Estimand	Effect of L on R	Direct and Indirect Effect of L on R	Effect of M on R
Direct effect of L on R	No	Yes	No
Unobs conf between L and M	No	No	No
Unobs conf between M and R	No	No	Yes
All variables observed together	No	Yes	Yes

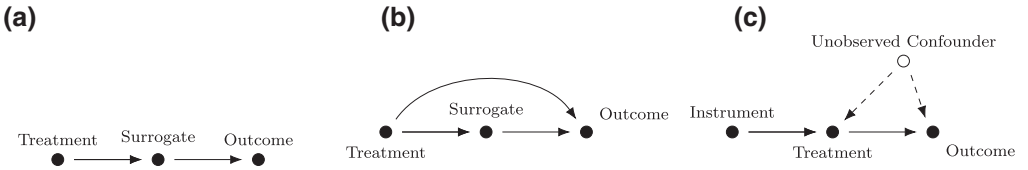


FIGURE 2
(A) Surrogacy assumption satisfied. (B) Violation of surrogacy due to direct effect (mediation setup) (C) Violation of surrogacy assumption due to unobserved confounder (IV setup)

3.4.1. Directed acyclical graph representations. The surrogacy, mediation, and instrumental variables literatures all study causal structures involving a causally linked sequence of three (sets) of variables. They differ in three key aspects: (1) the assumptions they make on the causal structure, (2) the estimands that are the primary focus of the analysis, and (3) the data available for the analyses. The literatures also differ in the labels typically used for the three variables. In Table 3, we list the labels, estimands, and some of the assumptions.

In Figure 2A–C, we show the differences in structures in DAG form in a single sample setting (so that we need not be concerned with the comparability assumption). Figure 2A illustrates the surrogacy setup, with a causal link from the treatment to the surrogate and from the surrogate to the outcome. There is no unobserved confounder for the causal relation between treatment and surrogate, which would violate Assumption 2 (Unconfounded Treatment Assignment/Strong Ignorability). There is no direct causal link from the treatment to the outcome. There are also no unobserved confounders for the causal relation between the surrogate and the outcome. These two features of the DAG (no direct link between treatment and outcome and no unobserved confounder for the relation between surrogate and outcome imply Assumption 3 (Surrogacy).

Figure 2B shows a mediation example where Assumption 3 is violated because there is a direct effect of the treatment on the outcome that does not pass through the surrogate. In this case, S_i is typically labelled as a mediator, rather than a surrogate. In the mediation case, the direct effect of the treatment on the outcome is estimable because all three variables, treatment, mediator, and outcome are observed in the same sample.

Figure 2C shows a DAG representation of the standard instrumental variables (IV) model familiar to economists. The first difference from the surrogacy setup in Figure 2A is that in the instrumental variables setting, the interest is in the causal effect of the variable in the middle of the three variable chain (the surrogate S in the surrogacy setting, and the treatment W in the instrumental variables setting), on the outcome, whereas in the surrogacy setting the primary interest is in the effect of the first variable in the chain (the treatment W in the surrogacy setting and the instrument Z in the instrumental variables setting) on the outcome. In the instrumental

Downloaded from academic.oup.com/restud/article/93/4/2284/8268796 by guest on 06 November 2026

variables case, the surrogacy estimand is immediately identified as the intention-to-treat effect of the instrument, since the instrument and the surrogate are observed in the same sample. Under the assumptions of the surrogacy setup, the target for an instrumental variables analysis, the effect of the surrogate on the primary outcome, is immediately identified. The instrumental variables setting is characterized by the presence of an unobserved confounder that affects both the treatment of interest and the outcome. The presence of that unobserved confounder violates Surrogacy, even if the treatment has no direct effect on the long-term outcome (Rosenbaum, 1984; Frangakis and Rubin, 2002; Joffe and Greene, 2009; VanderWeele, 2015).

The presence of this unobserved confounder also violates the comparability assumption if the marginal distribution of the treatment W differs between the observational and experimental samples, as will typically be the case. In both the surrogacy and the instrumental variables cases, we assume the absence of a direct effect of the first variable in the causal chain (the treatment W in the surrogacy case and the instrument in the instrumental variables case) on the primary outcome. In the surrogacy setting, this assumption is part of the Surrogacy assumption, while in the instrumental variables setting, this is typically referred to as the exclusion restriction (Angrist *et al.*, 1996).

3.4.2. A missing data representation. In the Online Appendix, we also discuss a missing data interpretation of the surrogacy approach. Essentially, we show that the following joint conditional independence assumption,

$$P_i \perp\!\!\!\perp Y_i \perp\!\!\!\perp W_i \mid S_i, X_i, \tag{3}$$

implies both surrogacy and comparability.

This missing data characterization is useful because it allows one to use insights from the missing data literature, both for the current problems and for generalizations. Given (3) we can use the conditional distribution of Y_i given (S_i, X_i) in the observational sample with $P_i = \text{O}$ to impute the missing outcomes in the experimental sample with $P_i = \text{E}$, and we can use the conditional distribution of W_i given (S_i, X_i) in the experimental sample with $P_i = \text{E}$ to impute the missing treatments in the observational sample with $P_i = \text{O}$.

This observation directly extends to more general imputation problems. Suppose we have two samples where in one sample, indicated by $P_i = \text{E}$ we observe one set of variables, (Z_{i1}, Z_{i2}) and in the second sample, indicated by $P_i = \text{O}$ we observe a partially overlapping set of variables, (Z_{i2}, Z_{i3}). Then, the analogous assumption that allows the imputation of all missing variables is $P_i \perp\!\!\!\perp Z_{i1} \perp\!\!\!\perp Z_{i3} \mid Z_{i2}$.

4. IDENTIFICATION AND SEMIPARAMETRIC EFFICIENCY BOUNDS

4.1. *Three identification results*

We now present our central identification result. We analyse three different representations of the ATE that lead to three estimation strategies, somewhat similar to inverse propensity score weighting, regression, and influence function estimators for ATEs under unconfoundedness (Imbens, 2004). The motivation for developing the different representations is that estimators corresponding to those different representations can have different properties in finite samples, just like they do in the unconfoundedness setting. Estimators based on the first representation require estimation of the surrogate index, but not the surrogate score. Estimators based on the second representation instead require estimation of the surrogate score, but not the surrogate index. Estimators based on the third representation require estimation of both, but have attractive double robustness properties.

Downloaded from academic.oup.com/restud/article/93/4/2284/8268796 by guest on 06 November 2026

We define the following four objects, all functionals of distributions that are directly estimable from the data. First define the statistical estimand, the average difference in the surrogate index between treated and control, adjusted for pre-treatment variables, in the experimental sample:

$$\begin{aligned}\tau^* \equiv \mathbb{E}\left[\{\mathbb{E}\left[\mathbb{E}\left[Y_i \middle| S_i, X_i, P_i = \mathrm{O}\right] \middle| W_i = 1, X_i, P_i = \mathrm{E}\right]\right. \\ \left. -\mathbb{E}\left[\mathbb{E}\left[Y_i \middle| S_i, X_i, P_i = \mathrm{O}\right] \middle| W_i = 0, X_i, P_i = \mathrm{E}\right]\} \middle| P_i = \mathrm{E}\right].\end{aligned} \tag{4}$$

Next, with a surrogate index representation:

$$\tau^{\mathrm{E}} \equiv \mathbb{E}\left[\mu(S_i, X_i, \mathrm{O}) \cdot \frac{W_i}{\rho(X_i)} - \mu(S_i, X_i, \mathrm{O}) \cdot \frac{1 - W_i}{1 - \rho(X_i)} \middle| P_i = \mathrm{E}\right], \tag{5}$$

then a surrogate score representation,

$$\begin{aligned}\tau^{\mathrm{O}} \equiv \mathbb{E}\left[Y_i \cdot \frac{\rho(S_i, X_i) \cdot \varphi(S_i, X_i) \cdot (1 - \varphi)}{\rho(X_i) \cdot (1 - \varphi(S_i, X_i)) \cdot \varphi}\right. \\ \left. - Y_i \cdot \frac{(1 - \rho(S_i, X_i)) \cdot \varphi(S_i, X_i) \cdot (1 - \varphi)}{(1 - \rho(X_i)) \cdot (1 - \varphi(S_i, X_i)) \cdot \varphi} \middle| P_i = \mathrm{O}\right].\end{aligned} \tag{6}$$

The third representation is based on the influence function. We first define

$$\mu(w, x) \equiv \mathbb{E}[\mu(S_i, X_i, \mathrm{O}) \mid W_i = w, X_i = x, P_i = \mathrm{E}].$$

Then, the influence function is

$$\begin{aligned}\psi(y, s, w, x, p) = \frac{\mathbf{1}_{p=\mathrm{E}}}{\varphi}\left(\frac{w \cdot (\mu(s, x, \mathrm{O}) - \mu(1, x))}{\rho(x)} - \frac{(1 - w) \cdot (\mu(s, x, \mathrm{O}) - \mu(0, x))}{1 - \rho(x)}\right) \\ + \frac{\mathbf{1}_{p=\mathrm{E}}}{\varphi}(\mu(1, x) - \mu(0, x) - \tau) \\ + \frac{\mathbf{1}_{p=\mathrm{O}}}{\varphi}\frac{\varphi(s, x)}{1 - \varphi(s, x)}\frac{(y - \mu(s, x, \mathrm{O}))\,(\rho(s, x) - \rho(x))}{\rho(x)(1 - \rho(x))}\end{aligned} \tag{7}$$

with the estimand

$$\tau^{\mathrm{O,E}} = \mathbb{E}\left[\psi(Y_i, S_i, W_i, X_i, P_i) + \tau\right]. \tag{8}$$

Remark 4. An earlier version of the paper had a mistake in the representation of the influence function. We are grateful to Kevin Chen and David Ritzwoller for pointing this out. See Chen and Ritzwoller (2023) for details.

Theorem 1 (Identification).

(i) *Suppose that Assumption 1 holds. Then, assuming all expectations are finite,*

$$\tau^* = \tau^{\mathrm{E}} = \tau^{\mathrm{O}} = \tau^{\mathrm{O,E}}.$$

(ii) *Suppose that Assumptions 1–4 hold. Then, the ATE is equal to the following three estimable functions of the data:*

$$\tau \equiv \mathbb{E}[Y_i(1) - Y_i(0) \mid P_i = \mathrm{E}] = \tau^* = \tau^{\mathrm{E}} = \tau^{\mathrm{O}} = \tau^{\mathrm{O,E}}.$$

Downloaded from academic.oup.com/restud/article/93/4/2284/8268796 by guest on 06 November 2026

(iii) *Jointly Assumptions 1, 2(i), 3(i), and 4(i) have no testable implications.*

Remark 5. The first part of the theorem implies that the four functionals of the joint distribution of $(\mathbf{1}_{P_i=\mathrm{O}}Y_i, S_i, \mathbf{1}_{P_i=\mathrm{E}}W_i, X_i, P_i)$ are identical, irrespective of the Unconfoundedness, Surrogacy, and Comparability assumptions.

Remark 6. Just like in the unconfoundedness case (Newey, 1994; Chernozhukov *et al.*, 2016), the influence function representation is doubly robust. Chen and Ritzwoller (2023) show that if the functions in the influence function that represent conditional expectations of the outcome, $\mu(s, x, \mathrm{O})$ and $\mu(w, x)$, are correctly specified, then the influence function has expectation zero irrespective of the functions used for the various propensity score, $\rho(s, x)$, $\rho(x)$, ρ, and the sampling score $\varphi(s, x)$. Similarly, if the various propensity score, $\rho(s, x)$, $\rho(x)$, ρ, and the sampling score $\varphi(s, x)$ are correct, the influence function has expectation zero, irrespective of the functions used for the conditional outcome expectations $\mu(s, x, \mathrm{O})$ and $\mu(w, x)$.

4.2. *Semiparametric efficiency bounds*

In this subsection, we present two pairs of semiparametric efficiency bound results (Newey, 1990; Bickel *et al.*, 1993) for two different data configurations. The first directly refers to the main setup in this paper with the experimental and observational sample. This result is essentially shown in Chen and Ritzwoller (2023), which corrects a mistake in an earlier version of the current paper.

Theorem 2. *Suppose Assumptions 1–4 hold. Then,*

(i) *the semiparametric efficiency bound, normalized by the square root of the sample size N, is*

$$\begin{aligned}\mathbb{V} &= \mathbb{E}[\psi(Y_i, S_i, W_i, X_i, P_i)^2] \\ &= \mathbb{E}\left[\frac{1-\varphi(S_i, X_i)}{\varphi^2}\left(\left(\frac{\varphi(S_i, X_i)}{1-\varphi(S_i, X_i)}\frac{\rho(S_i, X_i)-\rho(X_i)}{\rho(X_i)(1-\rho(X_i))}\right)^2\sigma^2(S_i, X_i, \mathrm{O})\right)\right. \\ &+ \frac{\varphi(X_i)}{\varphi^2}(\mu(1, X_i)-\mu(0, X_i)-\tau)^2 \\ &+ \frac{\varphi(S_i, X_i)}{\varphi^2}\left(\frac{(1-\rho(S_i, X_i))(\mu(S_i, X_i, \mathrm{O})-\mu(0, X_i))^2}{(1-\rho(X_i))^2}\right. \\ &\left.\left.+ \frac{\rho(S_i, X_i)(\mu(S_i, X_i, \mathrm{O})-\mu(1, X_i))^2}{\rho(X_i)^2}\right)\right].\end{aligned}$$

(ii) *If, in addition, the observational sample is large relative to the experimental sample, and* $\sup_{s,x}\varphi(s, x) \to 0$, *then the efficiency bound, now normalized by the expected sample size of the experimental sample,* $\mathbb{E}[N_{\mathrm{E}}] = \varphi N$ *simplifies to*

$$\begin{aligned}&\mathbb{E}\left[(\mu(1, X_i)-\mu(0, X_i)-\tau)^2 + \frac{(1-W_i)(\mu(S_i, X_i, \mathrm{O})-\mu(0, X_i))^2}{(1-\rho(X_i))^2}\right. \\ &\left.+\frac{W_i(\mu(S_i, X_i, \mathrm{O})-\mu(1, X_i))^2}{\rho(X_i)^2}\,\middle|\, P_i = \mathrm{E}\right].\end{aligned}$$

Downloaded from academic.oup.com/restud/article/93/4/2284/8268796 by guest on 06 November 2026

Remark 7. This variance in part (ii) of Theorem 2 is smaller than the efficiency bound we would obtain in a randomized experiment where we do observe the primary outcome but do not observe the surrogate. The bound in that case is well known since Hahn (1998),

$$\mathbb{E}\left[(\mu(1,X_i)-\mu(0,X_i)-\tau)^2+\frac{(1-W_i)(Y_i-\mu(0,X_i))^2}{(1-\rho(X_i))^2}+\frac{W_i(Y_i-\mu(1,X_i))^2}{\rho(X_i)^2}\middle| P_i=\mathrm{E}\right].$$

This advantage in terms of asymptotic precision of using the (true) predicted outcome $\mu(S_i,X_i,\mathrm{O})$ rather than the actual outcome Y_i has been noted previously in Day and Duffy (1996) in a setting with binary outcomes. In the general case, this gain is equal to

$$\mathbb{E}\left[\frac{(1-W_i)(Y_i-\mu(S_i,X_i,\mathrm{O}))^2}{(1-\rho(X_i))^2}+\frac{W_i(Y_i-\mu(S_i,X_i,\mathrm{O}))^2}{\rho(X_i)^2}\middle| P_i=\mathrm{E}\right].$$

Next, we consider the case where in a single sample we observe the treatment, primary outcome, surrogates and pre-treatment variables. In this single sample case, we do not need the fifth variable, $P_i\in\{\mathrm{E},\mathrm{O}\}$. To maintain consistency with the other parts of the discussion and to avoid ambiguity, we keep the notation as before. In this case, we can think of P_i always taking the value $P_i=\mathrm{E}$. We calculate the efficiency bound both without the assumption that surrogacy holds and with the assumption that surrogacy holds. We do so for a data generating process where surrogacy does hold, to see the information gain from that assumption.

Theorem 3. *Suppose Assumptions 2 and 3 hold.*

(i) *The variance bound without assuming surrogacy is*

$$\begin{aligned}\mathbb{V}_{\mathrm{ns}}=\mathbb{E}\Bigg[&\sigma^2(S_i,X_i,\mathrm{E})\cdot\left(\frac{\rho(S_i,X_i)}{\rho(X_i)^2}+\frac{1-\rho(S_i,X_i)}{(1-\rho(X_i))^2}\right)+(\mu(1,X_i)-\mu(0,X_i)-\tau)^2\\&+\frac{\rho(S_i,X_i)}{\rho(X_i)^2}\cdot(\mu(S_i,X_i,\mathrm{E})-\mu(1,X_i))^2\\&+\frac{1-\rho(S_i,X_i)}{(1-\rho(X_i))^2}\cdot(\mu(S_i,X_i,\mathrm{E})-\mu(0,X_i))^2\Bigg].\end{aligned}$$

(ii) *The efficiency gain from assuming surrogacy is*

$$\Delta=\mathbb{V}_{\mathrm{ns}}-\mathbb{V}_{\mathrm{s}}=\mathbb{E}\left[\sigma^2(S_i,X_i,E)\frac{\rho(S_i,X_i)(1-\rho(S_i,X_i))}{\rho(X_i)^2(1-\rho(X_i))^2}\right]\geq 0,$$

where $\mathbb{V}_{\mathrm{s}}$ is the variance bound for the case with surrogacy,

$$\begin{aligned}\mathbb{V}_{\mathrm{s}}=\mathbb{E}\Bigg[&\sigma^2(S_i,X_i,\mathrm{E})\left(\frac{\rho(S_i,X_i)-\rho(X_i)}{\rho(X_i)(1-\rho(X_i))}\right)^2+(\mu(1,X_i)-\mu(0,X_i)-\tau)^2\\&+\frac{\rho(S_i,X_i)}{\rho(X_i)^2}(\mu(S_i,X_i,\mathrm{E})-\mu(1,X_i))^2\\&+\frac{1-\rho(S_i,X_i)}{(1-\rho(X_i))^2}(\mu(S_i,X_i,\mathrm{E})-\mu(0,X_i))^2\Bigg].\end{aligned}$$

Downloaded from academic.oup.com/restud/article/93/4/2284/8268796 by guest on 06 November 2026

Remark 8. The expression for $\mathbb{V}_{\text{ns}}$ is equivalent to the efficiency bound in Hahn (1998). It is written here in terms of the surrogates to facilitate the comparison to the efficiency bound exploiting surrogacy.

Remark 9. Note that the variance bound in part (ii) of Theorem 3 differs from that in Theorem 2(ii) which was derived under the same surrogacy assumption, but assuming that the observational sample was infinitely large, so the relation between the surrogates and the primary outcome was known without error. The result in (ii) captures just the value of the surrogacy assumption.

5. VIOLATIONS OF THE SURROGACY AND COMPARABILITY ASSUMPTIONS: BIASES AND BOUNDS

The three critical assumptions, Unconfoundedness, Surrogacy, and Comparability, are strong. There is a large literature studying the sensitivity to unconfoundedness conditions (Rosenbaum and Rubin, 1983a; Imbens, 2003; Cinelli and Hazlett, 2020) or bounds (Manski, 1990). Multiple studies have also raised concerns that in practice Surrogacy may not be satisfied (Rosenbaum, 1984; Freedman *et al.*, 1992; Begg and Leung, 2000; Frangakis and Rubin, 2002; Joffe and Greene, 2009; VanderWeele, 2015), although we are not aware of formal sensitivity or bounds analyses. Violations of Comparability have not been explored because this assumption has not been previously formalized. In this section, we examine the biases that arise from violations of Surrogacy and Comparability. We first characterize these biases and then derive estimable bounds on the magnitude of the biases that can arise from such violations.

5.1. *Biases*

We begin by characterizing the probability limit of estimators based on the representations of the estimand, τ^{E}, τ^{O}, and $\tau^{\text{O,E}}$, in Theorem 1 when the Surrogacy and Comparability assumptions are violated, as well as in cases where the surrogate index is misspecified. Throughout the section, we maintain Unconfoundedness in the experimental sample (Assumption 2), and the random sampling assumption (Assumption 1). We denote the probability limit of the estimators by $\underline{\tau}$ to differentiate it from the ATE $\tau = \mathbb{E}[Y_i(1) - Y_i(0) \mid P_i = \text{E}]$.

Theorem 4.

(i) *Suppose Assumption 2 (Unconfoundedness) holds, but Assumptions 3(Surrogacy) and 4(Comparability) do not necessarily hold. Then*

$$\underline{\tau} \equiv \tau^{\text{O}} = \tau^{\text{E}} = \tau^{\text{E,O}} = \mathbb{E}\left[\,\mu(S_i(1), X_i, \text{O}) - \mu(S_i(0), X_i, \text{O})|\, P_i = \text{E}\right].$$

(ii) *Suppose Assumptions 2 (Unconfoundedness) and 4 (Comparability) hold, but Assumption 3 (Surrogacy) does not necessarily hold. Then, the difference between the average causal effect and the estimand is*

$$\textit{(surrogacy-bias)} \quad \tau - \underline{\tau} = \mathbb{E}\left[\{\mu(S_i, 1, X_i, \text{E}) - \mu(S_i, 0, X_i, \text{E})\} \times \frac{\rho(S_i, X_i) \cdot (1 - \rho(S_i, X_i))}{\rho(X_i) \cdot (1 - \rho(X_i))} \,\middle|\, P_i = \text{E}\right].$$

(iii) *Suppose Assumptions 2 (Unconfoundedness) and 3 (Surrogacy) hold, but Assumption 4 (Comparability) does not necessarily hold. Then the difference between the average causal effect and the estimand is*

Downloaded from academic.oup.com/restud/article/93/4/2284/8268796 by guest on 06 November 2026

$$\text{(comparability-bias)} \quad \tau - \underline{\tau} = \mathbb{E}\left[\{\mu(S_i, X_i, \mathrm{E}) - \mu(S_i, X_i, \mathrm{O})\} \times \frac{\rho(S_i, X_i) - \rho(X_i)}{\rho(X_i) \cdot (1 - \rho(X_i))} \,\middle|\, P_i = \mathrm{E}\right].$$

(*iv*). *Suppose Assumption 2 (Unconfoundedness) holds, but Assumptions 3 (Surrogacy) and 4 (Comparability) do not necessarily hold. Then, the difference between the average causal effect and the estimand is*

$$\begin{aligned}\textit{(total bias)} \quad \tau - \underline{\tau} = {} & \mathbb{E}\left[(\mu(S_i, 1, X_i, \mathrm{E}) - \mu(S_i, 0, X_i, \mathrm{E})) \times \frac{(1 - \rho(S_i, X_i)) \cdot \rho(S_i, X_i)}{(1 - \rho(X_i)) \cdot \rho(X_i)} \mid P_i = \mathrm{E}\right] \\ & + \mathbb{E}\left[(\mu(S_i, X_i, \mathrm{E}) - \mu(S_i, X_i, \mathrm{O})) \times \frac{\rho(S_i, X_i) - \rho(X_i)}{(1 - \rho(X_i)) \cdot \rho(X_i)} \mid P_i = \mathrm{E}\right].\end{aligned}$$

Remark 10. Theorem 4(*i*) shows that even without Surrogacy and Comparability, we estimate a valid average causal effect as long as unconfoundedness holds. The treatment effect we estimate is the average effect of the treatment on the surrogate index—a principled aggregate of intermediate outcomes—rather than the average effect on the primary outcome. This result also shows that the interpretation does not change with the choice of estimator (using the surrogate score approach, the surrogate index approach, or the influence function). Theorem 4(*ii–iv*) show how violations of Comparability or Surrogacy affect the difference between what is being estimated and the ATE on the primary outcome.

Remark 11. The bias from violations of Surrogacy (Theorem 4(*ii*)) consists of two factors. The first factor is small if the treatment does not explain much of the variation in Y_i and therefore $\mu(s, 1, x, \mathrm{E})$ and $\mu(s, 0, x, \mathrm{E})$ are close. The second factor is small if the surrogate explains a large share of the variation in W_i, so that the surrogate score is close to zero or one and therefore $\mathbb{E}[\rho(S_i, X_i) \cdot (1 - \rho(S_i, X_i))]$ is close to zero.

Remark 12. The bias from violations of Comparability (Theorem 4(*iii*)) also consists of two factors. The first is the difference between the surrogacy index $\mu(s, x, \mathrm{O})$ and its counterpart in the experimental sample, $\mu(s, x, \mathrm{E})$. The second factor depends on the deviation between the surrogacy score and the propensity score, $\rho(S_i, X_i) - \rho(X_i)$. If the treatment does not have much effect on the surrogates, violations of Comparability do not generate much bias, because the bias that comes from a combination of the effect of the treatment on the surrogates and the effect of the surrogates on the outcome, will be small in that case.

5.2. *Bounds on the bias*

In this subsection, we explore bounds on the parameter of interest. We show that, in general, these bounds are uninformative. However, if outcomes themselves are bounded, for example, if the outcomes are binary, informative bounds can be derived. Moreover, we present bounds given assumptions on the range of violations of the Surrogacy and Comparability assumptions.

Downloaded from academic.oup.com/restud/article/93/4/2284/8268796 by guest on 06 November 2026

Lemma 1. *Suppose Assumptions 2 (Unconfoundedness) and 4 (Comparability) hold, but Assumption 3 (Surrogacy) does not necessarily hold. Then:*

(i) *If the outcome can take on values on the whole real line, then there is no value for the ATE τ that can be ruled out.*

(ii) *If the outcome is binary, then the ATE τ is inside the interval*

$$\begin{aligned}\{&\mathbb{E}\left[\mu(S_i(1), X_i, \mathrm{O}) - \mu(S_i(0), X_i, \mathrm{O})\middle| P_i = \mathrm{E}\right] \\ &+\mathbb{E}\left[\Delta_S^L(S_i, X_i)\frac{\rho(S_i, X_i)(1-\rho(S_i, X_i))}{\rho(X_i)(1-\rho(X_i))}\middle| P_i = \mathrm{E}\right], \\ &\mathbb{E}\left[\mu(S_i(1), X_i, \mathrm{O}) - \mu(S_i(0), X_i, \mathrm{O})\middle| P_i = \mathrm{E}\right] \\ &+\mathbb{E}\left[\Delta_S^U(S_i, P_i)\frac{\rho(S_i, X_i)(1-\rho(S_i, X_i))}{\rho(X_i)(1-\rho(X_i))}\middle| P_i = \mathrm{E}\right]\Bigg\},\end{aligned}$$

where

$$\begin{aligned}\Delta_S^L(s, x) &= -\min\left(\frac{1-\mu(s, x, \mathrm{O})}{\rho(s, x)}, \frac{\mu(s, x, \mathrm{O})}{1-\rho(s, x)}\right) \\ \Delta_S^U(s, x) &= \min\left(\frac{\mu(s, x, \mathrm{O})}{\rho(s, x)}, \frac{1-\mu(s, x, \mathrm{O})}{1-\rho(s, x)}\right),\end{aligned}$$

and this bound is sharp.

(iii) *If the direct effect of the treatment on the outcome $\mu(S_i, 1, X_i, \mathrm{E}) - \mu(S_i, 0, X_i, \mathrm{E})$ is bounded in absolute value by c, then the ATE τ is inside the interval*

$$\begin{aligned}\{&\mathbb{E}\left[\mu(S_i(1), X_i, \mathrm{O}) - \mu(S_i(0), X_i, \mathrm{O})\middle| P_i = \mathrm{E}\right] \\ &- c\cdot\mathbb{E}\left[\frac{\rho(S_i, X_i)(1-\rho(S_i, X_i))}{\rho(X_i)(1-\rho(X_i))}\middle| P_i = \mathrm{E}\right], \\ &\mathbb{E}\left[\mu(S_i(1), X_i, \mathrm{O}) - \mu(S_i(0), X_i, \mathrm{O})\middle| P_i = \mathrm{E}\right] \\ &+ c\cdot\mathbb{E}\left[\frac{\rho(S_i, X_i)(1-\rho(S_i, X_i))}{\rho(X_i)(1-\rho(X_i))}\middle| P_i = \mathrm{E}\right]\Bigg\},\end{aligned}$$

and this bound is sharp.

Remark 13. To provide some intuition for the sharpness of the bounds, consider the surrogacy bias in Theorem 4. The bias has two factors, with the second estimable from the data. The first factor is the difference $\mu(S_i, 1, X_i, \mathrm{E}) - \mu(S_i, 0, X_i, \mathrm{E})$. The data are not directly informative about this difference beyond the fact that the weighted average $\rho(S_i, X_i)\mu(S_i, 1, X_i, \mathrm{E}) + (1 - \rho(S_i, X_i))\mu(S_i, 0, X_i, \mathrm{E})$ is equal to the estimable quantity $\mu(S_i, X_i, \mathrm{O})$. In the absence of any restrictions on the outcome, this implies there are no restrictions on $\mu(S_i, w, X_i, \mathrm{E})$ or on the difference $\mu(S_i, 1, X_i, \mathrm{E}) - \mu(S_i, 0, X_i, \mathrm{E})$, and thus not on the bias or the ATE. Given restrictions on the range of the outcome, this representation directly leads to upper and lower bounds on the bias and the ATE.

Lemma 2. *Suppose Assumptions 2 (Unconfoundedness) and 3 (surrogacy) hold, but Assumption 4 (Comparability) does not necessarily hold. Then:*

(i) *If the outcome can take on values on the whole real line, then there is no value for the ATE τ that can be ruled out.*

Downloaded from academic.oup.com/restud/article/93/4/2284/8268796 by guest on 06 November 2026

(ii) *If the outcome is binary, then the ATE τ is inside the interval*

$$\Bigg\{\mathbb{E}\left[\mu(S_i(1), X_i, \mathrm{O}) - \mu(S_i(0), X_i, \mathrm{O})\middle| P_i = \mathrm{E}\right] + \mathbb{E}\left[\left\{\mathbf{1}_{\rho(S_i,X_i)<\rho(X_i)} - \mu(S_i, X_i, \mathrm{O})\right\}\frac{\rho(S_i, X_i) - \rho(X_i)}{\rho(X_i)(1-\rho(X_i))}\middle| P_i = \mathrm{E}\right],$$
$$\mathbb{E}\left[\mu(S_i(1), X_i, \mathrm{O}) - \mu(S_i(0), X_i, \mathrm{O})\middle| P_i = \mathrm{E}\right] + \mathbb{E}\left[\left\{\mathbf{1}_{\rho(S_i,X_i)>\rho(X_i)} - \mu(S_i, X_i, \mathrm{O})\right\}\frac{\rho(S_i, X_i) - \rho(X_i)}{\rho(X_i)(1-\rho(X_i))}\middle| P_i = \mathrm{E}\right]\Bigg\},$$

with width

$$2\mathbb{E}\left[\mathbf{1}_{\rho(S_i,X_i)>\rho(X_i)}\cdot\frac{\rho(S_i, X_i) - \rho(X_i)}{\rho(X_i)\cdot(1-\rho(X_i))}\middle| P_i = \mathrm{E}\right],$$

and this bound is sharp.

(iii) *If $\mu(S_i, X_i, \mathrm{E}) - \mu(S_i, X_i, \mathrm{O})$ is bounded in absolute value by c, then the ATE τ is inside the interval*

$$\Bigg\{\mathbb{E}\left[\mu(S_i(1), X_i, \mathrm{O}) - \mu(S_i(0), X_i, \mathrm{O})\middle| P_i = \mathrm{E}\right] - c\cdot\mathbb{E}\left[\frac{|\rho(S_i, X_i) - \rho(X_i)|}{\rho(X_i)\cdot(1-\rho(X_i))}\middle| P_i = \mathrm{E}\right],$$
$$\mathbb{E}\left[\mu(S_i(1), X_i, \mathrm{O}) - \mu(S_i(0), X_i, \mathrm{O})\middle| P_i = \mathrm{E}\right] + c\cdot\mathbb{E}\left[\frac{|\rho(S_i, X_i) - \rho(X_i)|}{\rho(X_i)\cdot(1-\rho(X_i))}\middle| P_i = \mathrm{E}\right]\Bigg\},$$

and this bound is sharp.

6. ESTIMATION

In this section, we first present four estimators for the ATE. The first, the surrogate index estimator, is related to previously proposed estimators, with the difference that in the earlier literature, the surrogate index was implicitly assumed to be known. We then discuss three new alternative estimators. The last of these new estimators is a matching estimator. Although matching estimators are generally not efficient in settings with unconfoundedness (Abadie and Imbens, 2006, 2016; Rubin, 2006), they are widely applied, and it is instructive to see how a matching strategy can be used here.

6.1. *Surrogate index*

Suppose we estimate the surrogate index as $\hat{\mu}(s, x, \mathrm{O})$ and the propensity score as $\hat{\rho}_{\mathrm{E}}(x)$. We take an average of the surrogate index in the experimental sample for the treatment and control

Downloaded from academic.oup.com/restud/article/93/4/2284/8268796 by guest on 06 November 2026

groups, after adjusting for the propensity score. A natural estimator, corresponding to (5), is the following difference of the two averages over the experimental sample:

$$\hat{\tau}^{\mathrm{E}} = \frac{1}{\sum_{i=1}^{N_{\mathrm{E}}} W_i/\hat{\rho}(X_i)} \sum_{i=1}^{N_{\mathrm{E}}} \hat{\mu}(S_i, X_i, \mathrm{O}) \cdot \frac{W_i}{\hat{\rho}(X_i)} - \frac{1}{\sum_{i=1}^{N_{\mathrm{E}}} (1-W_i)/(1-\hat{\rho}(X_i))} \sum_{i=1}^{N_{\mathrm{E}}} \hat{\mu}(S_i, X_i, \mathrm{O}) \cdot \frac{1-W_i}{1-\hat{\rho}(X_i)}. \tag{9}$$

We refer to this as the surrogate index estimator. Note that compared to the representation in Theorem 1, we normalize the weights so that the weights sum up to one. This tends to improve the finite sample properties of related estimators in other settings substantially (Hirano *et al.*, 2003; Busso *et al.*, 2014).

In the case where the estimator for the surrogate index $\mu(s, x, \mathrm{O})$ was based on a linear specification for the regression of the primary outcome on the intermediate outcome, $\mu(s, x, \mathrm{O}) = \gamma_0 + \gamma_S' s + \gamma_X' x$, this leads to

$$\hat{\tau}^{\mathrm{E}} = \hat{\gamma}_S' \hat{\tau}_S,$$

where $\hat{\tau}_S$ is an estimator for the average effect of the treatment on the surrogates, $\mathbb{E}[S_i(1) - S_i(0)]$. In the simplest case, without pre-treatment variables and where the experimental sample is randomized, $\hat{\tau}_S = \overline{S}_1 - \overline{S}_0$, where $\overline{S}_1$ and $\overline{S}_0$ are the average values of the surrogate outcomes. Here, the estimator simplifies to the difference in the estimated surrogate index in the treatment group and the control group: $\hat{\tau}^{\mathrm{E}} = \hat{\gamma}_S'(\overline{S}_1 - \overline{S}_0)$. This expression is also familiar from the mediation literature (*e.g.* Baron and Kenny, 1986) and the surrogacy literature (Day and Duffy, 1996). However, we emphasize that in general there may be interactions between the surrogates and pre-treatment variables, or between some of the surrogates. In that case the linear specification need not be not adequate.

6.2. *Surrogate score estimator*

We now use the second representation for τ in the main theorem to derive an alternative estimator. Let $\hat{\rho}(x)$, $\hat{\rho}(s, x)$, $\hat{\varphi}(s, x)$, $\hat{\varphi}(x)$, and $\hat{\varphi}$, be estimators for $\rho(x)$, $\rho(s, x)$, $\varphi(s, x)$, $\varphi(x)$, and φ, respectively.

The surrogate score estimator is based on averaging the following expression over the observational sample:

$$\hat{\tau}^{\mathrm{O}} = \frac{1}{\sum_{i \mid P_i=\mathrm{O}} \omega_{1,i}} \sum_{i \mid P_i=\mathrm{O}} Y_i \cdot \omega_{1,i} - \frac{1}{\sum_{i \mid P_i=\mathrm{O}} \omega_{0,i}} \sum_{i \mid P_i=\mathrm{O}} Y_i \cdot \omega_{0,i}, \tag{10}$$

where for $w = 0, 1$ the weights are

$$\omega_{w,i} = \frac{\hat{\rho}(S_i, X_i)^w \cdot (1-\hat{\rho}(S_i, X_i))^{1-w} \cdot \hat{\varphi}(S_i, X_i) \cdot (1-\hat{\varphi})}{\hat{\rho}(X_i)^w \cdot (1-\hat{\rho}(X_i))^{1-w} \cdot (1-\hat{\varphi}(S_i, X_i)) \cdot \hat{\varphi}}. \tag{11}$$

6.3. *Influence function estimator*

We can also base estimation on the efficient score given in (7). Given estimators for the propensity score, the surrogate score, and the sampling score, we can estimate the average treatment

Downloaded from academic.oup.com/restud/article/93/4/2284/8268796 by guest on 06 November 2026

effect as

$$\hat{\tau}^{E,O} = \sum_{i=1}^{N} \left\{ \frac{\mathbf{1}_{P_i=\mathrm{E}}}{\hat{\varphi}} \left(\frac{W_i \cdot \hat{\mu}(S_i, X_i, \mathrm{O})}{\hat{\rho}(X_i)} - \frac{(1-W_i) \cdot \hat{\mu}(S_i, X_i, \mathrm{O})}{1-\hat{\rho}(X_i)} \right) \right.$$
$$+ \frac{\mathbf{1}_{P_i=\mathrm{E}}}{\hat{\varphi}} \left(\hat{\mu}(1, X_i) \left(1 - \frac{W_i}{\hat{\rho}(X_i)} \right) - \hat{\mu}(0, X_i) \left(1 - \frac{1-W_i}{1-\hat{\rho}(X_i)} \right) \right)$$
$$\left. + \frac{\mathbf{1}_{P_i=\mathrm{O}}}{1-\hat{\varphi}} \left(\frac{\hat{\varphi}(S_i, X_i)}{1-\hat{\varphi}(S_i, X_i)} \frac{1-\hat{\varphi}}{\hat{\varphi}} \right) \frac{(Y_i - \hat{\mu}(S_i, X_i, \mathrm{O}))\left(\hat{\rho}(S_i, X_i) - \hat{\rho}(X_i)\right)}{\hat{\rho}(X_i)(1-\hat{\rho}(X_i))} \right\}. \quad (12)$$

Based on the results in Newey (1994), it follows that under standard conditions the two estimators above and the surrogate index estimator all reach the semiparametric efficiency bound, and are first-order equivalent.

The recent literature on double robust estimation of ATEs under unconfoundedness (Chernozhukov *et al.*, 2016) suggests that this estimator may have superior properties in small samples.

6.4. *Double-matching estimator*

Consider unit i in the experimental sample with $X_i = x$ and $S_i = s$, and suppose this is a treated unit with $W_i = 1$. We need to find three matches for this unit. First, we need to find a unit with the opposite treatment in the same (experimental) sample. Specifically, we need to find the closest unit in the experimental sample, in terms of pre-treatment variables, among the units with $W_i = 0$. Suppose this unit is unit j, with $W_j = 0$, and the value of the pre-treatment variables for this unit are $X_j = x'$, and the surrogate outcomes are $S_j = s'$. As a result of the matching, we should have $x \approx x'$, but potentially s could be quite different from s'. Next, we need to find for each of the two units i and j a match in the observational sample. Find the unit in the observational sample closest to unit i, in terms of both pre-treatment variables and surrogates. Let i' be the index for this unit, and let the value of the outcome for this unit be $Y_{i'}$, and the values of the pre-treatment variables and surrogates $X_{i'}$ and $S_{i'}$. Now, as a result of the matching $X_i \approx X_{i'}$ and $S_i \approx S_{i'}$. Finally, find the unit in the observational sample closest to unit j, in terms of both pre-treatment variables and surrogates. Let the value of the outcome for this unit be $Y_{j'}$, and the values of the pre-treatment variables and surrogates $X_{j'}$ and $S_{j'}$, with $X_j \approx X_{j'}$ and $S_j \approx S_{j'}$.

Then, we combine these matches to estimate the causal effect for unit i, $Y_i(1) - Y_i(0)$, as the difference in average outcomes for the two matches from the observational sample:

$$\widehat{Y_i(1) - Y_i(0)} = Y_{i'} - Y_{j'}. \quad (13)$$

The matching estimator for τ would then be the average value of (13) over the experimental sample. The double-matching estimator is then

$$\hat{\tau}^{\mathrm{match}} = \frac{1}{N^{\mathrm{E}}} \sum_{i:P_i\mathrm{E}} \left\{ W_i \left(Y_{i'} - Y_{j'} \right) + (1 - W_i) \left(Y_{j'} - Y_{i'} \right) \right\}.$$

7. APPLICATION: IMPACTS OF JOB TRAINING ON EMPLOYMENT

In this section, we apply our method to estimate the causal effect of the GAIN job training programme on long-term labour market outcomes. GAIN was a job assistance programme

Downloaded from academic.oup.com/restud/article/93/4/2284/8268796 by guest on 06 November 2026

implemented in California in the 1980s to help welfare recipients find work (Riccio *et al.*, 1989; Friedlander and Robins, 1995; Hotz *et al.*, 2006). The Manpower Demonstration Research Corporation conducted a randomized trial to evaluate the GAIN programme's employment impacts in six counties in California in the late 1980s. We focus primarily on the GAIN trial in Riverside, which was widely heralded as the programme that had the largest treatment effects on earnings. The Riverside programme emphasized a "jobs first" approach to re-entry into the labour force, encouraging unemployed workers to take any job they find; in contrast, other sites focused more heavily on developing human capital through training programmes (Hotz *et al.*, 2006).

We have available long-term outcomes for the four GAIN sites, including employment, earnings, and receipt of aid over the first thirty-six quarters after random assignment. We take the average of the thirty-six employment indicators and earnings in Riverside as our primary outcomes. We then investigate whether we could have predicted the long-term impact on these outcomes using only the first T quarters of all outcomes (including employment, earnings, and aid) as surrogates, as well as using pre-treatment variables (characteristics of the individuals as well as lagged employment, earnings and aid outcomes). The Riverside data on the treatment, surrogates and pre-treatment variables play the role of our experimental sample. We use the data from the combination of the other three locations (Alameda, Los Angeles, and San Diego) as our observational sample. For the observational sample, we only use the information on the surrogates, pre-treatment variables, and outcome, but not the treatment assignment, nor the indicator for the location.

We begin by presenting a brief summary of the samples. We then describe how we construct our surrogate index. Next, we illustrate our theoretical results by evaluating the magnitude of the gains from using surrogate indices in terms of time and precision relative to existing experimental estimates of the programme's long-term impacts in Riverside. We also show how one can validate the surrogacy assumption using intermediate outcomes and bound the degree of bias arising from potential violations of surrogacy.

7.1. *The GAIN programme*

The GAIN treatment was randomly assigned to welfare (Aid for Families with Dependent Children) recipients, a very low-income population. The treatment group consisted of $N_{\mathrm{E},T} = 4{,}405$ participants, which the control group consisted of $N_{\mathrm{E},C} = 1{,}040$ participants who were not eligible for the additional services in the GAIN programme. The data we use come from the Hotz *et al.* (2006), which followed study participants for 9 years after assignment of the treatment, measuring quarterly employment rates and earnings[2] from the Unemployment Insurance database. They found that the treatment effects of the Riverside GAIN programme on employment rates and earnings were initially large, but declined over time, as shown in Figure 3A, which plots employment rates by quarter for individuals in the experimental (Riverside) treatment and control groups, and in Figure 3B, which shows the corresponding results for quarterly earnings.

In Riverside, the estimated causal effects on the primary outcomes were a 6.4 (s.e. = 1.2) percentage point (pp) increase in average quarterly employment rates, and an \$249 (s.e. \$84) increase in average quarterly earnings, in both cases averaged over the thirty-six quarter post-treatment. Our question is whether these impacts could have been estimated more quickly by using short-term employment, earnings and aid receipt as surrogates.

2. All income variables were converted to 1999 dollars using cost-of-living deflators; see footnote 21 of Hotz *et al.* (2006) for more information.

Downloaded from academic.oup.com/restud/article/93/4/2284/8268796 by guest on 06 November 2026

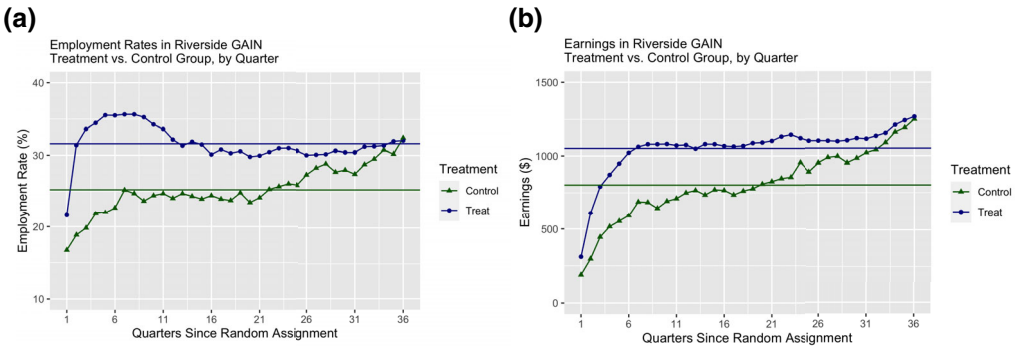


FIGURE 3

Outcomes by quarter since assignment for treatment and control. (A) Employment. (B) Earnings

The observational sample includes the other three locations, Alameda, Los Angeles, and San Diego, for a total of $N_{\rm O} = 13,725$ individuals.

In the online appendix Table 8 presents information on the pre-treatment variables. Clearly, the two samples, Riverside and the combination of the other three locations, are substantially different prior to the intervention in terms of permanent characteristics such as ethnicity, as well as in pre-treatment outcomes.

7.2. *Three estimators*

We discuss here the estimators for the average effect of the programme. We wish to consider different set of surrogates, indexed by the number of periods t we want to use as surrogates. To capture this, we index the surrogate for individual i, S_i^t, by the superscript t. S_i^t contains the employment indicators, earnings outcomes and aid receipt indicators for the t quarters after the intervention.

7.2.1. Surrogate index estimator. To construct the surrogacy index, we estimate a linear regression model using least squares, for the individuals in the observational sample

$$Y_i = \beta_0 + \beta_S^\top S_i^t + \beta_X^\top X_i + \varepsilon_i. \tag{14}$$

The predicted value from this regression, which we denote by $\hat{Y}_i$, is our surrogate index for mean employment based on surrogates up to quarter t. We then compute this surrogate index for each of the individuals in the experimental sample and estimate the treatment effect based on the surrogate index as

$$\hat{\tau}^{\rm O} = \frac{1}{N_{{\rm E},T}} \sum_{i=1}^{N^{\rm E}} \hat{Y}_i W_i - \frac{1}{N_{{\rm E},C}} \sum_{i=1}^{N_{\rm E}} \hat{Y}_i (1 - W_i). \tag{15}$$

If we use all thirty-six quarters of employment indicators as surrogates, then the regression of Y_i on the set of surrogates will fit perfectly, $\hat{Y}_i$ will be equal to Y_i, and the estimated effect will be identical to the original experimental estimate. The question is whether using a much more limited set of surrogates will get us close to the experimental benchmark.

7.2.2. Surrogate score estimator. For the surrogate score estimator, we first estimate a logistic regression of the treatment indicator on the pre-treatment variables and the surrogates. We specify

Downloaded from academic.oup.com/restud/article/93/4/2284/8268796 by guest on 06 November 2026

$$\ln\left(\frac{\rho(S_i^t, X_i)}{1-\rho(S_i^t, X_i)}\right) \equiv \ln\left(\frac{\text{pr}(W_i = 1 \mid S_i^t, X_i, P_i = \text{E})}{1-\text{pr}(W_i = 1 \mid S_i^t, X_i, P_i = \text{E})}\right) = \alpha_0 + \alpha_S^\top S_i^t + \alpha_X^\top X_i,$$

and estimate this on the experimental (Riverside) sample.

Next, we estimate the propensity score, also using logistic regression,

$$\ln\left(\frac{\rho(X_i)}{1-\rho(X_i)}\right) \equiv \ln\left(\frac{\text{pr}(W_i = 1 \mid X_i, P_i = \text{E})}{1-\text{pr}(W_i = 1 \mid X_i, P_i = \text{E})}\right) = \delta_0 + \delta_X^\top X_i,$$

and estimate this again on the experimental (Riverside) sample. In principle, the random assignment implies that the δ_X should be close to zero in this case.

Finally, we estimate the comparability score

$$\ln\left(\frac{\varphi(S_i^t, X_i)}{1-\varphi(S_i^t, X_i)}\right) \equiv \ln\left(\frac{\text{pr}(P_i = \text{E} \mid X_i, S_i^t)}{1-\text{pr}(P_i = \text{E} \mid X_i, S_i^t)}\right) = \gamma_0 + \gamma_S^\top S_i^t + \gamma_X^\top X_i,$$

and estimate this on the combined observational and experimental samples.

The surrogate score estimator is based on averaging the following expression over the observational sample:

$$\hat{\tau}^{\text{O}} = \frac{1}{\sum_{i \mid P_i=\text{O}} \omega_{1,i}} \sum_{i \mid P_i=\text{O}} Y_i \cdot \omega_{1,i} - \frac{1}{\sum_{i \mid P_i=\text{O}} \omega_{0,i}} \sum_{i \mid P_i=\text{O}} Y_i \cdot \omega_{0,i}, \tag{16}$$

where the weights are as before in equation (11).

7.2.3. Influence function estimator. For the influence function estimator, we first estimate the surrogacy index, the surrogacy score, the propensity score, and the comparability score as before. We then plug those into the estimator in equation (12).

7.3. *Results*

Here, we discuss two sets of results. First, the estimates for the average effect of the intervention on the two primary outcomes under various assumptions about the surrogates. Second, we test the Surrogacy and Comparability assumptions directly.

7.3.1. Estimation results. As the discussion after the surrogate index estimator shows, we recover the experimental estimates if we use all thirty-six quarters of employment indicators as surrogates. The question is whether we can do approximately as well with fewer than 36 quarters of surrogates. In Figure 4A and B, we compare the experimental estimates of the effect on the primary outcomes (0.064 for the employment outcome, and $249 for the earnings outcome) to the three sets of surrogate estimates, as a function of how many periods of surrogates we use, ranging from one quarter to thirty-six quarters. To put this in perspective, we also include in these two figures what we label the "naive" estimator, where we estimate the effect on the long-term outcome as the effect on the first t quarters of the outcome. In Tables 4 and 5, we report a subset of the numbers underlying these estimates with the corresponding standard errors.

We see that the naive estimator does very poorly. It takes more than twenty-five quarters before the naive estimator is within two standard errors of the experimental estimate. In contrast, all three surrogate-based estimators are all within two standard errors when the surrogates include five quarters of outcomes, for both outcomes.

Downloaded from academic.oup.com/restud/article/93/4/2284/8268796 by guest on 06 November 2026

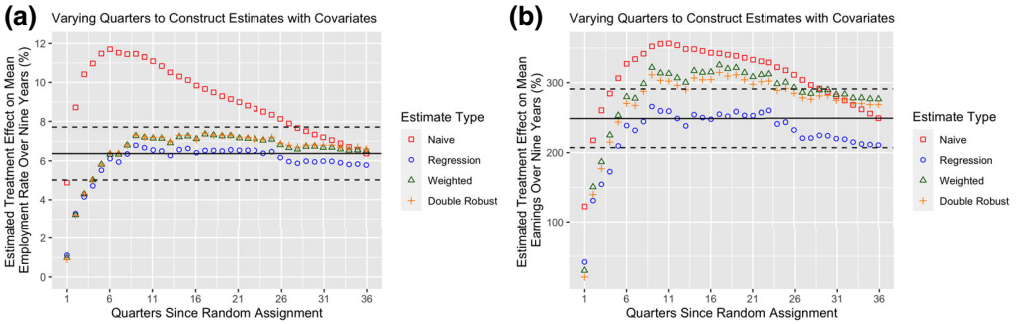


FIGURE 4

Sensitivity to choices of surrogates

TABLE 4

Estimates for effect on employment

	Naive		Surrogacy index		Surrogacy score		Influence function	
t	est	(s.e.)	est	(s.e.)	est	(s.e.)	est	(s.e.)
1	0.049	(0.013)	0.011	(0.003)	0.010	(0.002)	0.010	(0.002)
2	0.087	(0.012)	0.033	(0.003)	0.032	(0.003)	0.033	(0.004)
3	0.104	(0.011)	0.042	(0.004)	0.043	(0.004)	0.044	(0.004)
4	0.110	(0.011)	0.047	(0.005)	0.050	(0.005)	0.052	(0.005)
5	0.115	(0.011)	0.055	(0.005)	0.058	(0.005)	0.060	(0.005)
6	0.117	(0.010)	0.061	(0.006)	0.063	(0.006)	0.065	(0.006)
12	0.108	(0.010)	0.065	(0.007)	0.071	(0.008)	0.073	(0.008)
18	0.095	(0.010)	0.065	(0.008)	0.073	(0.009)	0.075	(0.009)
24	0.085	(0.010)	0.064	(0.009)	0.070	(0.010)	0.072	(0.010)
30	0.073	(0.010)	0.059	(0.009)	0.067	(0.010)	0.070	(0.010)
36	0.064	(0.010)	0.058	(0.009)	0.065	(0.010)	0.068	(0.010)

Experimental benchmark: 0.064 (s.e. 0.012).

TABLE 5

Estimates for effect on earnings

	Naive		Surrogacy index		Surrogacy score		Influence function	
t	est	(s.e.)	est	(s.e.)	est	(s.e.)	est	(s.e.)
1	122.4	(28.9)	41.8	(13.4)	29.6	(9.5)	31.0	(10.4)
2	217.5	(30.1)	131.1	(18.3)	150.7	(18.9)	150.4	(19.1)
3	260.6	(31.8)	154.5	(23.4)	186.8	(24.1)	187.5	(24.0)
4	284.4	(33.5)	172.7	(27.0)	225.1	(28.0)	225.1	(27.6)
5	306.5	(35.2)	209.6	(29.5)	253.4	(30.3)	254.4	(30.0)
6	327.1	(36.6)	238.8	(31.5)	279.7	(32.5)	280.6	(32.3)
12	353.9	(41.3)	249.1	(39.4)	306.8	(42.9)	308.6	(42.9)
18	340.2	(43.9)	252.3	(44.3)	320.1	(45.7)	321.8	(45.9)
24	322.2	(46.5)	241.0	(49.8)	298.3	(49.7)	300.8	(50.2)
30	286.5	(48.5)	224.3	(50.3)	289.9	(50.5)	293.0	(51.1)
36	249.1	(50.0)	210.9	(50.2)	276.6	(51.4)	279.6	(52.1)

Experimental benchmark: $249 (s.e. $83).

7.3.2. Validation results and other supplementary analyses. Given the data available, we can also test whether using t quarters of surrogates is sufficient to satisfy Surrogacy and Comparability. To test Surrogacy, we regress the primary outcome on the pre-treatment variables,

Downloaded from academic.oup.com/restud/article/93/4/2284/8268796 by guest on 06 November 2026

TABLE 6
Surrogacy and comparability assumption tests for employment Outcome

	Surrogacy assumption			Comparability assumption		
t	est	(s.e)	T-stat	est	(s.e)	T-stat
1	0.052	(0.010)	5.4	0.008	(0.005)	1.7
2	0.034	(0.009)	3.7	−0.004	(0.004)	−0.9
3	0.024	(0.009)	2.6	−0.006	(0.004)	−1.5
4	0.018	(0.009)	2.0	−0.007	(0.004)	−1.8
5	0.010	(0.008)	1.2	−0.010	(0.004)	−2.5
6	0.004	(0.008)	0.5	−0.011	(0.004)	−3.0
12	−0.004	(0.006)	−0.7	−0.015	(0.003)	−5.0
18	−0.007	(0.004)	−1.6	−0.009	(0.002)	−4.1
24	−0.005	(0.003)	−1.8	−0.005	(0.001)	−3.6
30	−0.002	(0.001)	−1.3	−0.001	(0.001)	−1.9
35	0.000	(0.000)	−2.0	0.000	(0.000)	−0.8

TABLE 7
Surrogacy and comparability assumption tests for earnings outcome

	Surrogacy assumption			Comparability assumption		
t	est	(s.e)	T-stat	est	(s.e)	T-stat
1	185.6	(50.8)	3.7	−35.6	(25.3)	−1.4
2	129.7	(49.9)	2.6	−65.0	(24.5)	−2.7
3	94.3	(48.1)	2.0	−72.8	(23.5)	−3.1
4	66.3	(46.3)	1.4	−67.5	(22.4)	−3.0
5	42.2	(44.5)	1.0	−71.0	(21.5)	−3.3
6	12.6	(42.0)	0.3	−73.9	(20.5)	−3.6
12	−19.1	(31.3)	−0.6	−65.2	(15.3)	−4.2
18	−41.8	(22.0)	−1.9	−31.2	(10.8)	−2.9
24	−20.2	(13.6)	−1.5	−27.4	(6.7)	−4.1
30	−10.8	(5.9)	−1.8	−4.7	(2.8)	−1.6
35	−0.5	(1.0)	−0.5	−0.4	(0.5)	−0.8

the surrogates up to quarter t, and the indicator for the treatment; a finding that the treatment has an impact indicates a violation of Surrogacy. We estimate this regression using a logistic regression model, using only the data from the experimental (Riverside) sample. We report in Tables 6 and 7 the results from these regressions for a number of different values for t, for the employment outcome and the earnings outcome. We report the point estimate, standard error, and t-statistic. We see that point estimates for $t \leq 3$ are large and highly statistically significant. After that, most of the t-statistics are less than 2, although there are some where the t-statistics are a little above 2, but the coefficient estimates are small.

We do a similar exercise for Comparability. We combine the experimental and observational samples and regress the final outcome on the surrogates, the pre-treatment variables, and an indicator for the experimental sample, again using surrogates up to period t. We report the estimates on the indicator for the experimental sample, and the corresponding standard error. Here, the point estimates become smaller after $t = 12$, but the t-statistics remain large even with a substantial number of surrogate periods, indicating a violation of Surrogacy.

If we are unwilling to make the surrogacy assumption, we can still calculate bounds for the effect on employment, using the fact that this outcome is binary. For the case where the first six quarters of post-treatment data are used as surrogates, the lower and upper bounds are estimated as −0.186 and 0.124. These are not very informative, because the data now do not allow us to estimate the indirect effect of the treatment on the outcome.

Downloaded from academic.oup.com/restud/article/93/4/2284/8268796 by guest on 06 November 2026

Similarly, we can calculate bounds for the average effect without assuming Comparability. With six quarters of surrogates, the bounds are again wide at −0.076 and 0.194, respectively. Here, the fact that the treatment effect on the surrogates is strong leads to substantial sensitivity to the comparability assumption as formalized in Lemma 2.

Using the data for Riverside, we can also assess the value of the Surrogacy assumption. Using the six quarters of data as surrogates, we find that the gain from knowledge of Surrogacy (the Δ in Theorem 3) is quite large. The standard error given Surrogacy, $\sqrt{\mathbb{V}_s}$, is 0.33 times the standard error without knowledge that Surrogacy holds, $\sqrt{\mathbb{V}_{ns}}$.

8. CONCLUSION

We develop new methods for combining intermediate outcomes to estimate the long-term impacts of treatments more rapidly and precisely. Our method requires estimating a "surrogate index"—the conditional expectation of the long-term outcome given intermediate outcomes—and then estimating the treatment effect on the surrogate index. The surrogate index can be estimated using parametric or nonparametric regression methods. We formalize conditions under which this method yields unbiased estimates, derive bounds for the degree of bias when those assumptions fail, and propose a simple out-of-sample validation approach using "hold out" intermediate outcomes. We show that surrogates can also greatly improve the precision of estimates even in settings where the treatment effect on the long-term outcome can be estimated directly, particularly when that outcome is rare or noisy.

Applying the method to analyse the impacts of the GAIN job training programme in California, we find that using short-term earnings and employment rates to construct surrogate indices expedite the detection of long-term treatment effects on employment and earnings by several years and also substantially increases precision. Furthermore, a single surrogate index accurately predicts heterogeneity in the long-term treatment effects of different types of job training programmes across sites, showing that surrogate indices estimated in a given setting may be generalizable to other settings. The success of the surrogate index in this application validates the use of short-term employment outcomes as surrogates for detecting longer-term impacts of job training programmes, an empirical result that can be applied when analysing ongoing programmes.

Building on this application, it would be useful to systematically establish surrogate indices that match the long-term treatment effects estimated in other experiments and quasi-experiments. Over time, this would allow researchers to collectively build a public library of surrogate indices for long-term outcomes that could be used to expedite the analysis of future interventions.

Acknowledgments. We are grateful for discussions with James Dailey, Lawrence Katz, David Ritzwoller, Dylan Small, Scott Stern, and Liang Xu, and for comments from numerous seminar participants. We thank Kevin Chen, Yechan Park, Emanuel Schertz, and James Stratton for outstanding research assistance. We are particularly grateful to Kevin Chen and David Ritzwoller for pointing out a mistake in the previous version of this paper. This research was funded through National Science Foundation Grant DMS-1502437, the Chan-Zuckerberg Initiative, the Bill & Melinda Gates Foundation, the Overdeck Foundation, ONR grants N00014-17-1-2131 and N00014-19-1-2468, and the Sloan Foundation.

Supplementary Data

Supplementary data are available at *Review of Economic Studies* online.

Data Availability

The code underlying this research is available on Zenodo at https://doi.org/10.5281/zenodo.13732512. The dataset we use (Data/quarterly.mat) comes from unpublished replication materials accompanying (Hotz *et al.*, 2006). We received

Downloaded from academic.oup.com/restud/article/93/4/2284/8268796 by guest on 06 November 2026

the dataset directly from the authors and have permission to use it for our analysis. No other dataset is necessary to complete our analysis. Researchers interested in getting access to these data can contact the authors of Hotz *et al.* (2006).

REFERENCES

ABADIE, A. and IMBENS, G. W. (2006), "Large Sample Properties of Matching Estimators for Average Treatment Effects", *Econometrica: Journal of the Econometric Society*, **74**, 235–267.

—— —— (2016), "Matching on the Estimated Propensity Score", *Econometrica: Journal of the Econometric Society*, **84**, 781–807.

ALONSO, A., MOLENBERGHS, G., GEYS, H., *et al.* (2006), "A Unifying Approach for Surrogate Marker Validation Based on Prentice's Criteria", *Statistics in Medicine*, **25**, 205–221.

ANGRIST, J. D., IMBENS, G. W. and RUBIN, D. B. (1996), "Identification of Causal Effects Using Instrumental Variables", *Journal of the American Statistical Association*, **91**, 444–455.

ATHEY, S., CHETTY, R. and IMBENS, G. (2020), 'Combining experimental and observational data to estimate treatment effects on long-term outcomes', arXiv, arXiv:2006.09676, preprint: not peer reviewed. https://doi.org/10.48550/arXiv.2006.09676

BARON, R. M. and KENNY, D. A. (1986), "The Moderator–Mediator Variable Distinction in Social Psychological Research: Conceptual, Strategic, and Statistical Considerations", *Journal of Personality and Social Psychology*, **51**, 1173–1182.

BEGG, C. B. and LEUNG, D. H. Y. (2000), "On the Use of Surrogate End Points in Randomized Trials", *Journal of the Royal Statistical Society: Series A (Statistics in Society)*, **163**, 15–28.

BICKEL, P. J., KLAASSEN, C. A. J., BICKEL, P. J., *et al.* (1993), *Efficient and Adaptive Estimation for Semiparametric Models* (Vol. **4**) (Baltimore: Johns Hopkins University Press).

BUSSO, M., DINARDO, J. and MCCRARY, J. (2014), "New Evidence on the Finite Sample Properties of Propensity Score Reweighting and Matching Estimators", *Review of Economics and Statistics*, **96**, 885–897.

CHEN, J. and RITZWOLLER, D. M. (2023), "Semiparametric Estimation of Long-Term Treatment Effects", *Journal of Econometrics*, **237**, 105545.

CHEN, X., HONG, H. and TAROZZI, A. (2008), "Semiparametric Efficiency in GMM Models with Auxiliary Data", *Annals of Statistics*, **36**, 808–843.

CHERNOZHUKOV, V., CHETVERIKOV, D., DEMIRER, M., *et al.* (2016), "Double Machine Learning for Treatment and Causal Parameters." Centre for Microdata Methods and Practice, Institute for Fiscal Studies.

CHETTY, R., FRIEDMAN, J. N., HILGER, N., *et al.* (2011), "How does Your Kindergarten Classroom Affect Your Earnings? Evidence from Project STAR", *The Quarterly Journal of Economics*, **126**, 1593–1660.

CINELLI, C. and HAZLETT, C. (2020), "Making Sense of Sensitivity: Extending Omitted Variable Bias", *Journal of the Royal Statistical Society: Series B, Statistical Methodology*, **82**, 39–67.

D'AGOSTINO, R. B., CAMPBELL, M. J. and GREENHOUSE, J. B. (2006), "Surrogate Markers: Back to the Future", *Statistics in Medicine*, **25**, 181–182.

DAY, N. E. and DUFFY, S. W. (1996), "Trial Design Based on Surrogate end Points—Application to Comparison of Different Breast Screening Frequencies", *Journal of the Royal Statistical Society: Series A (Statistics in Society)*, **159**, 49–60.

FLEMING, T. R. and DEMETS, D. L. (1996), "Surrogate end Points in Clinical Trials: Are We Being Misled?", *Annals of Internal Medicine*, **125**, 605–613.

FRANGAKIS, C. E. and RUBIN, D. B. (2002), "Principal Stratification in Causal Inference", *Biometrics*, **58**, 21–29.

FREEDMAN, L. S., GRAUBARD, B. I. and SCHATZKIN, A. (1992), "Statistical Validation of Intermediate Endpoints for Chronic Diseases", *Statistics in Medicine*, **11**, 167–178.

FRIEDLANDER, D. and ROBINS, P. K. (1995), "Evaluating Program Evaluations: New Evidence on Commonly Used Nonexperimental Methods", *The American Economic Review*, **85**, 923–937.

GILBERT, P. B. and HUDGENS, M. G. (2008), "Evaluating Candidate Principal Surrogate Endpoints", *Biometrics*, **64**, 1146–1154.

GRAHAM, B. S., DE XAVIER PINTO, C. C. and EGEL, D. (2016), "Efficient Estimation of Data Combination Models by the Method of Auxiliary-To-Study Tilting (AST)", *Journal of Business & Economic Statistics*, **34**, 288–301.

GUPTA, S., KOHAVI, R., TANG, D., *et al.* (2019), "Top Challenges from the First Practical Online Controlled Experiments Summit", *ACM SIGKDD Explorations Newsletter*, **21**, 20–35.

HAHN, J. (1998), "On the Role of the Propensity Score in Efficient Semiparametric Estimation of Average Treatment Effects", *Econometrica: Journal of the Econometric Society*, **66**, 315–331.

HECKMAN, J. J., STIXRUD, J. and URZUA, S. (2006), "The Effects of Cognitive and Noncognitive Abilities on Labor Market Outcomes and Social Behavior", *Journal of Labor Economics*, **24**, 411–482.

HIRANO, K., IMBENS, G. W. and RIDDER, G. (2003), "Efficient Estimation of Average Treatment Effects Using the Estimated Propensity Score", *Econometrica: Journal of the Econometric Society*, **71**, 1161–1189.

HOLLAND, P. W. (1986), "Statistics and Causal Inference", *Journal of the American Statistical Association*, **81**, 945–960.

HOTZ, V. J., IMBENS, G. W. and KLERMAN, J. A. (2006), "Evaluating the Differential Effects of Alternative Welfare to Work Training Components: A Reanalysis of the California GAIN Program", *Journal of Labor Economics*, **24**, 521–566.

Downloaded from academic.oup.com/restud/article/93/4/2284/8268796 by guest on 06 November 2026

—— —— —— (2006), "Evaluating the Differential Effects of Alternative Welfare-To-Work Training Components: A Reanalysis of the California GAIN Program", *Journal of Labor Economics*, **24**, 521–566.
HOTZ, V. J., IMBENS, G. W. and MORTIMER, J. H. (2005), "Predicting the Efficacy of Future Training Programs Using Past Experiences at Other Locations", *Journal of Econometrics*, **125**, 241–270.
IMAI, K., KEELE, L. and TINGLEY, D. (2010), "A General Approach to Causal Mediation Analysis", *Psychological Methods*, **15**, 309–334.
IMBENS, G. (2004), "Nonparametric Estimation of Average Treatment Effects Under Exogeneity: A Review", *Review of Economics and Statistics*, **86**, 4–29.
IMBENS, G. W. (2003), "Sensitivity to Exogeneity Assumptions in Program Evaluation", *The American Economic Review, Papers and Proceedings*, **93**, 126–132.
IMBENS, G. W. and RUBIN, D. B. (2015), *Causal Inference in Statistics, Social, and Biomedical Sciences* (Cambridge: Cambridge University Press).
JOFFE, M. M. and GREENE, T. (2009), "Related Causal Frameworks for Surrogate Outcomes", *Biometrics*, **65**, 530–538.
KALLUS, N. and MAO, X. (2025), "On the Role of Surrogates in the Efficient Estimation of Treatment Effects with Limited Outcome Data", *Journal of the Royal Statistical Society Series B: Statistical Methodology*, **87**, 480–509.
LALONDE, R. J. (1986), "Evaluating the Econometric Evaluations of Training Programs with Experimental Data", *The American Economic Review*, **76**, 604–620.
LAURITZEN, S. L. (2004), "Discussion on Causality", *Scandinavian Journal of Statistics*, **31**, 189–193.
LITTLE, R. J. A. and RUBIN, D. B. (2014), *Statistical Analysis With Missing Data* (Vol. **333**) (Hoboken, NJ: John Wiley & Sons).
MANSKI, C. F. (1990), "Nonparametric Bounds on Treatment Effects", *The American Economic Review*, **80**, 319–323.
NEWEY, W. K. (1990), "Semiparametric Efficiency Bounds", *Journal of Applied Econometrics*, **5**, 99–135.
—— (1994), "The Asymptotic Variance of Semiparametric Estimators", *Econometrica: Journal of the Econometric Society*, **62**, 1349–1382.
PEARL, J. (2000), *Causality: Models, Reasoning, and Inference* (New York, NY, USA: Cambridge University Press).
PEARL, J. and BAREINBOIM, E. (2014), "External Validity: From Do-Calculus to Transportability Across Populations", *Statistical Science*, **29**, 579–595.
PRENTICE, R. L. (1989), "Surrogate Endpoints in Clinical Trials: Definition and Operational Criteria", *Statistics in Medicine*, **8**, 431–440.
QU, Y. and CASE, M. (2006), "Quantifying the Indirect Treatment Effect via Surrogate Markers", *Statistics in Medicine*, **25**, 223–231.
RÄSSLER, S. (2004), "Data Fusion: Identification Problems, Validity, and Multiple Imputation", *Austrian Journal of Statistics*, **33**, 153–171.
—— (2012), *Statistical Matching: A Frequentist Theory, Practical Applications, and Alternative Bayesian Approaches* (**Vol. 168**) (New York, NY: Springer Science & Business Media).
RICCIO, J., GOLDMAN, B., HAMILTON, G., *et al.* (1989), "GAIN: Early Implementation Experiences and Lessons. California's Greater Avenues for Independence Program." (Memo).
RIDDER, G. and MOFFITT, R. (2007), "The Econometrics of Data Combination", *Handbook of Econometrics*, **6**, 5469–5547.
ROSENBAUM, P. R. (1984), "The Consequences of Adjustment for a Concomitant Variable that has Been Affected by the Treatment", *Journal of the Royal Statistical Society: Series A (General)*, **147**, 656–666.
ROSENBAUM, P. R. and RUBIN, D. B. (1983a), "Assessing Sensitivity to an Unobserved Binary Covariate in an Observational Study with Binary Outcome", *Journal of the Royal Statistical Society. Series B (Methodological)*, **45**, 212–218.
—— —— (1983b), "The Central Role of the Propensity Score in Observational Studies for Causal Effects", *Biometrika*, **70**, 41–55.
RUBIN, D. B. (1974), "Estimating Causal Effects of Treatments in Randomized and Nonrandomized Studies", *Journal of Educational Psychology*, **66**, 688–701.
—— (1976), "Inference and Missing Data", *Biometrika*, **63**, 581–592.
—— (2004), *Multiple Imputation for Nonresponse in Surveys* (Vol. **81**) (New York, NY: John Wiley & Sons).
—— (2006), *Matched Sampling for Causal Effects* (Cambridge: Cambridge University Press).
TCHETGEN TCHETGEN, E. J. and SHPITSER, I. (2014), "Estimation of a Semiparametric Natural Direct Effect Model Incorporating Baseline Covariates", *Biometrika*, **101**, 849–864.
VAN DER LAAN, M. J. and PETERSEN, M. L. (2004), "Estimation of Direct and Indirect Causal Effects in Longitudinal Studies." (Memo).
VANDERWEELE, T. (2015), *Explanation in Causal Inference: Methods for Mediation and Interaction* (Oxford: Oxford University Press).
WEIR, C. J. and WALLEY, R. J. (2006), "Statistical Evaluation of Biomarkers as Surrogate Endpoints: A Literature Review", *Statistics in Medicine*, **25**, 183–203.
XU, J. and ZEGER, S. L. (2001), "The Evaluation of Multiple Surrogate Endpoints", *Biometrics*, **57**, 81–87.
ZHENG, W. and VAN DER LAAN, M. J. (2012), "Targeted Maximum Likelihood Estimation of Natural Direct Effects", *The International Journal of Biostatistics*, **8**, 1–40.

Downloaded from academic.oup.com/restud/article/93/4/2284/8268796 by guest on 06 November 2026